

Why Do Firms Pay More for Bank Loans?

The Role of Renegotiation

Zhuolu Gao*

April 6, 2026

Abstract

Firms that borrow from both banks and the corporate bond market pay a substantial premium on bank loans. I show that a large share of this premium compensates banks for providing renegotiation flexibility in out-of-court restructurings. Exploiting a court ruling that made such restructurings more difficult, I find that more affected firms experienced an 80–90 bps decline in the loan premium, consistent with fewer restructuring opportunities and a reduced ability to avoid bankruptcy costs. These findings point to renegotiation flexibility as a central driver of the loan premium and highlight the distinct value of bank lending.

JEL classification codes: G21, G32, G33

Keywords: bank loan, corporate bond, loan premium, renegotiation, prepayment risk

*Frankfurt School of Finance & Management, z.gao@fs.de. I thank Samuel Antill (discussant), Denis Davydov (discussant), Jens Dick-Nielsen, Peter Feldhütter, Paul Fontanier, Janet Gao, Christopher James (discussant), Andrey Kurbatov (discussant), David Lando, Anton Lines, Lasse Heje Pedersen, Lorian Pelizzon, Oliver-Alexander Press, Yingjie Qi, Akash Raja, Natalia Rostova (discussant), Alessandro Spina, and Yao Zeng for helpful comments. I also thank seminar and conference participants at the NFN Workshop, SFI PhD Workshop, FIRS 2024, the Tri-City Day-Ahead Workshop, BIGFI Research Retreat, Erasmus School of Economics, Fudan University, Shanghai Advanced Institute of Finance, and WFA 2025. I gratefully acknowledge financial support from the Center for Financial Frictions (Grant No. DNR102), the Center for Big Data in Finance (Grant No. DNR167), and the European Research Council (ERC Grant No. 101040188).

1 Introduction

Firms borrow using two primary instruments, private bank loans and public bonds. One would expect firms to optimize their debt funding sources and choose the cheaper instrument if there were large spread differences. Yet [Schwert \(2020\)](#) shows that banks charge a substantially higher spread than the spread implied by the bond market, after adjusting for seniority. This difference, referred to as the ‘loan premium’, amounts to 140–170 bps and accounts for roughly half of the total loan spread. This raises questions about both the supply and demand dynamics of the loan market: why are banks able to charge such a premium, and why are firms willing to pay more for bank loans despite having access to cheaper bond financing?

This paper shows that a large share of the loan premium reflects the renegotiation flexibility banks provide in financial distress. Specifically, banks’ ability and willingness to facilitate out-of-court restructurings allow firms to avoid the substantial costs associated with formal bankruptcy procedures. Firms are therefore willing to pay more for bank debt because it provides not only funding, but also flexibility when repayment problems arise. I formalize this idea in a stylized model, where banks can support an out-of-court restructuring by extending debt maturity. The model highlights two implications for the empirical analysis. First, renegotiation flexibility helps explain why bank loans can command a premium over public debt. Second, a shock that raises creditor coordination frictions should reduce this premium more for firms that rely more heavily on bond financing.

To empirically test these implications, I first reproduce the loan premium following [Schwert \(2020\)](#). I construct a dataset of more than 10,000 syndicated loans to public firms from 1997 to 2025 and find that bank lenders charge a premium of 96 bps compared to the public bond

market.¹ I then study how this premium changes following an unexpected legal shock to the feasibility of out-of-court restructuring. In 2014, the Marblegate ruling raised concerns that distressed restructurings would become more difficult to implement without unanimous bondholder consent ([Court of the Southern District of New York, 2014](#)). Evidence in [Kornejew \(2025\)](#) shows that the ruling worsened holdout problems and pushed more distressed, bond-reliant firms into formal bankruptcy. This ruling provides an exogenous shock to the feasibility of out-of-court restructurings, making it an ideal setting to assess the effect on the loan premium.

Using a difference-in-differences design, I find that the loan premium falls by 80–90 basis points for bond-intensive firms after the Marblegate ruling. This decline suggests that when out-of-court restructuring becomes less feasible, the value of banks’ renegotiation flexibility declines as well. The effect is smaller for loans involving institutional lenders and larger for loans with a single lead arranger, consistent with the idea that renegotiation is easier when fewer parties are involved (e.g., [Demiroglu and James, 2015](#)). These results support the view that banks earn a substantial premium because they provide restructuring flexibility that public debt markets cannot easily replicate.

In addition to renegotiation flexibility, I show that a smaller portion of the loan premium compensates banks for the prepayment and repricing flexibility embedded in loan contracts. Building on [Roberts and Sufi \(2009\)](#) and [Roberts \(2015\)](#), I show in a stylized setting that compensation for prepayment risk should be higher for firms with a greater probability of future credit improvement. Consistent with this prediction, I find that a higher loan premium forecasts subsequent credit rating upgrades. The implied compensation for prepayment flexibility is economically meaningful but modest, around 20–30 basis points, suggesting that prepayment risk explains only a limited share of the overall loan premium. These findings also suggest that

¹When restricting the sample to term loans only, as in [Schwert \(2020\)](#), I find a loan premium of around 122–142 bps, which closely aligns with the original study’s findings.

standard loan fees do not fully compensate lenders for prepayment risk.²

Overall, the evidence shows that banks provide valuable flexibility beyond funding itself. The main source of this value is the ability to support distressed renegotiation and help firms avoid bankruptcy costs. A smaller component comes from allowing borrowers to refinance or reprice debt when conditions improve. Together, these results explain why firms pay a substantial premium for bank loans and highlight the distinct role of banks relative to public debt markets.

Related Literature

My paper contributes to the literature on banks' ability to create value for their borrowers. Banks perform valuable functions such as screening and monitoring (Leland and Pyle, 1977; Diamond, 1984; Ramakrishnan and Thakor, 1984; Fama, 1985), maturity and liquidity transformation, and risk diversification (Diamond and Dybvig, 1983), which can mitigate financial frictions. Another set of empirical papers provides indirect evidence of the value of bank lending. James (1987) shows that the stock market reaction to new loan announcements is positive. Berger and Udell (1995) find that borrowers with longer banking relationships pay lower interest rates and are less likely to pledge collateral. Bharath et al. (2011) also arrive at the conclusion that repeated borrowing from the same lender translates into a 10–17 bps lowering of loan spreads. Datta et al. (1999) find that the existence of bank debt lowers the spreads for first public straight bond offers by about 68 bps. In particular, Hoshi et al. (1990) present evidence that relationship banking can reduce the costs of financial distress using data on Japanese firms. The saved costs stem from the inherent difficulty of renegotiating financial claims, particularly

²In practice, over 90% of loans can be prepaid without any cancellation fee. While most syndicated loans include an upfront fee, the amount is often not disclosed to all syndicate participants or the public. Berg et al. (2016) use hand-collected SEC filings of syndicated loan contracts and show that many contracts refer to a nonpublic fee letter without disclosing the upfront fee. Since all syndicate members bear prepayment risk, compensating them through the interest spread rather than upfront fees may be more natural, especially because upfront fees are concentrated among lead arrangers.

when there are many dispersed creditors. [Gilson et al. \(1990\)](#) also document the pattern that distressed firms that owe more of their debt to banks are more likely to succeed in out-of-court restructuring and avoid the presumably more costly Chapter 11 bankruptcy. In this paper, I provide empirical evidence showing that banks charge an interest rate premium for the saved distress costs. However, in the event of a shock that significantly increases the difficulty of renegotiation and restructuring, the premium vanishes.

This paper also speaks to the bank loan contract design. [Gorton and Kahn \(2000\)](#) argue that the initial loan rate is not set to price the risk of default, but rather to minimize subsequent costs associated with moral hazard and renegotiation. As originally discussed in [Hart and Moore \(1988\)](#), debt contracts are inherently incomplete. [Roberts \(2015\)](#) finds that a typical bank loan is renegotiated five times throughout the loan life. The pricing, maturity, amount, and covenants are all significantly adjusted during each renegotiation. [Denis and Wang \(2014\)](#) find that even in the absence of covenant violations, there are frequent renegotiations, primarily relaxing existing restrictions and resulting in economically large changes in existing limits. Furthermore, [Chodorow-Reich and Falato \(2022\)](#) examine whether lenders' financial health can transmit to borrowers through the covenant violation channel. My paper confirms the pattern that the initial loan rate is subject to changes in conditions, rather than solely reflecting the credit risk.

More broadly, this paper relates to the research on debt structure and debtors' rights of firms. There are several explanations for the decision on the choices between public debt and private debt, such as the probability of inefficient liquidations, control of moral hazard problems, and the cost of disclosing proprietary information ([Diamond, 1984, 1991](#); [Chemmanur and Fulghieri, 1994](#); [Hackbarth et al., 2007](#); [De Fiore and Uhlig, 2011](#)). More recently, [Becker and Ivashina \(2014\)](#) empirically examine firms' debt choices between bank loans and public

bonds, whereas [Morellec et al. \(2015\)](#) build a model of investment and financing decisions to study the choice between bonds and loans. This line of research focuses more on the quantity of new issues of debt while this paper focuses more on the prices of the two sources of debt. There is an essential difference between bond credit and loan credit in that loan credit is more concentrated with a few creditors based on a relationship, whereas bond securities are issued in the capital market, where many dispersed investors can buy and sell small positions. The misaligned motivations between dispersed bondholders and relationship banks can be costly for firms. First, the dispersed creditors suffer from a collective action problem, which can create incentives to hold out the debt and free-ride on others' concessions. Second, the debtor and its relationship lenders might take advantage of restructuring debt strategically ([Gertner and Scharfstein, 1991](#); [Brudney, 1992](#); [Kornejew, 2025](#)). This paper provides indirect evidence on how the misaligned motivations due to dispersed creditors can destroy stakeholders' value.

2 Model

This section presents a stylized framework for why firms may pay more for bank loans than for bonds. The key idea is that bank debt embeds contractual flexibility: in distress, banks can renegotiate and help firms avoid inefficient liquidation. When firms also rely on public bonds, this flexibility becomes less effective because dispersed bondholders may hold out and make restructuring harder to complete. I model this mechanism first and then turn to the prepayment option embedded in bank loans.

2.1 Renegotiation in the Pure Bank-Debt Case

I begin with a reduced-form setup. Let ρ_0 denote the firm's probability of distress at date 1 and let α denote the loss given default (LGD). The face value of debt is normalized to 1 and the risk-free rate is set to zero, so the debt spread compensates lenders only for expected credit losses. Figure 1 illustrates the timing.

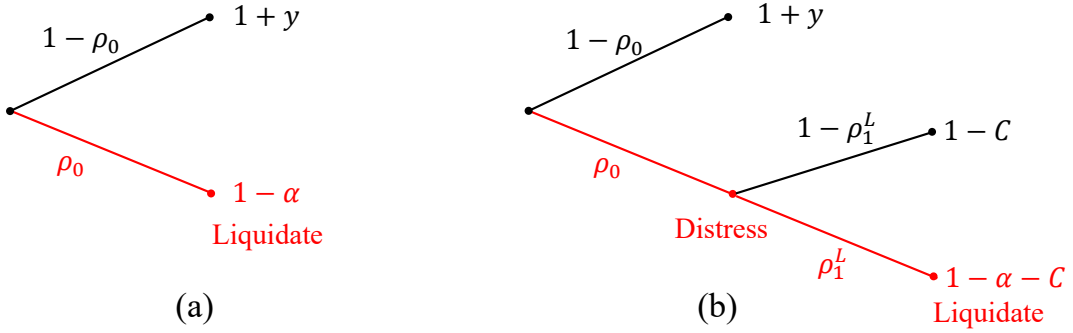


Figure 1: Distress resolution with and without renegotiation. In Panel (a), debt matures at date 1. The firm repays in full if solvent and is liquidated if distressed. In Panel (b), the bank can renegotiate in distress and extend the debt to date 2. Conditional on renegotiation, the firm continues and defaults at date 2 with probability ρ_1^L . The bank incurs a renegotiation cost C .

Benchmark debt. Without renegotiation, debt is priced to break even:

$$1 = (1 - \rho_0)(1 + y) + \rho_0(1 - \alpha), \quad (1)$$

which implies

$$y^* = \frac{\alpha \rho_0}{1 - \rho_0}. \quad (2)$$

This benchmark rate reflects only expected credit loss.

Bank debt with renegotiation. Suppose instead that a relationship bank can extend the loan in distress.³ Conditional on renegotiation, the borrower survives to date 2 and defaults there with probability ρ_1^L . The bank incurs renegotiation cost C .⁴ The break-even loan rate is then

$$1 = (1 - \rho_0)(1 + y) + \rho_0 [\rho_1^L(1 - \alpha) + (1 - \rho_1^L) - C], \quad (3)$$

so that

$$y_l^* = \frac{\alpha \rho_0 \rho_1^L}{1 - \rho_0} + \frac{\rho_0 C}{1 - \rho_0}. \quad (4)$$

To separate the benefit of renegotiation from its cost, define the lower bound on the renegotiated loan rate as

$$\underline{y}^* \equiv \frac{\alpha \rho_0 \rho_1^L}{1 - \rho_0},$$

which corresponds to the case of zero renegotiation cost ($C = 0$).

Relative to the benchmark rate y^* , renegotiation has two opposing effects. It lowers expected liquidation losses because the firm can continue in distress ($\underline{y}^* < y^*$), but it raises the rate through the renegotiation cost C . Now, the loan rate reflects both the benefit of avoiding liquidation and the cost of restructuring.

While the loan rate y_l^* reflects the lender's break-even condition, the borrower's willingness to accept such a contract depends on whether the continuation value from avoiding immediate liquidation is sufficiently large. Let V_2 denote firm value at date 2 in the recovery state. Appendix A.1 derives the renegotiation-feasibility condition by comparing shareholder payoffs

³I model renegotiation as a maturity extension for simplicity. In practice, out-of-court restructurings may also involve changes in coupons, principal reductions, or covenant adjustments. The key feature for the analysis is that renegotiation allows the firm to avoid immediate liquidation and continue operating.

⁴For simplicity, the liquidation loss parameter α is assumed to be the same after renegotiation. Allowing for further deterioration in asset value would not affect the qualitative results and can be interpreted as being captured by the renegotiation cost C . I also assume that if the firm recovers after renegotiation, the lender receives only principal at date 2. One can show that the main qualitative results are unchanged if the lender instead receives both principal and interest.

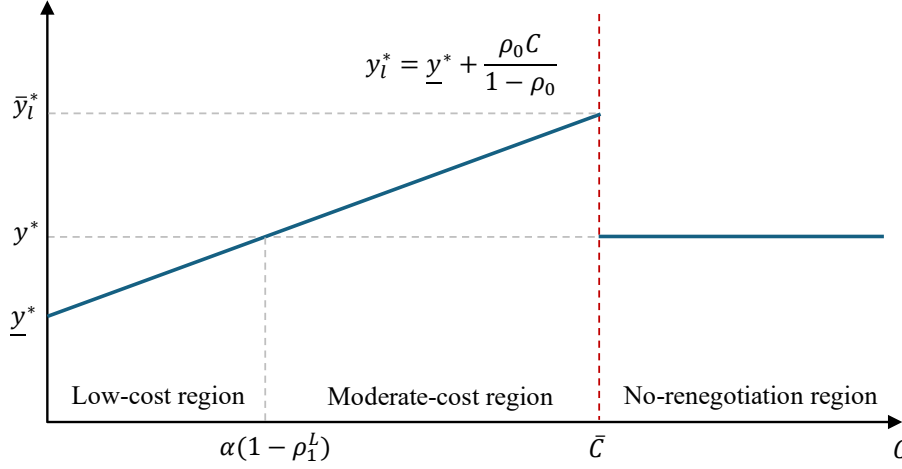


Figure 2: Loan rate and renegotiation cost. The figure plots the loan rate as a function of the renegotiation cost C . Renegotiation occurs only when $C < \bar{C}$. For low values of C , the reduction in expected liquidation losses dominates and the loan rate falls below the benchmark rate y^* . For higher values of C , the cost term dominates and the loan rate exceeds y^* . When $C \geq \bar{C}$, renegotiation is not feasible and the loan rate equals the benchmark.

under liquidation and continuation. Renegotiation is attractive if and only if

$$C < \bar{C} = (1 - \rho_1^L)(\alpha + V_2 - 1). \quad (5)$$

Combining the pricing equation with this threshold condition yields three regions. As illustrated in Figure 2, when C is sufficiently small, the reduction in expected losses dominates and the loan rate can fall below the benchmark rate. For larger values of C , the cost term dominates and the loan rate increases. When $C \geq \bar{C}$, renegotiation is not feasible and the loan rate reverts to the benchmark.

The threshold $\bar{C}$ summarizes the total surplus from renegotiation: the reduction in liquidation losses together with the continuation value accruing to shareholders.

Proposition 1. *In the pure bank-debt case, renegotiation is feasible only when its cost is below the total surplus it creates, $C < \bar{C}$. When this condition holds, the loan rate reflects two opposing forces: a reduction in expected liquidation losses and an increase due to renegotiation*

costs, leading to a non-monotonic relationship between the loan rate and C .

The pure bank case is useful for intuition, but it cannot generate a loan premium relative to bonds. That comparison requires mixed borrowing.

2.2 Renegotiation in the Mixed-Borrowing Case

I now move to the empirically relevant case in which the firm borrows from both a relationship bank and the corporate bond market. The senior bank loan has face value $1 - F_b$ and the junior bond has face value F_b , with $F_b < \alpha$. This assumption ensures that in liquidation the junior bond is fully wiped out, while the senior loan still bears default risk.

Without renegotiation, the firm is liquidated immediately in distress. In this case, the senior loan recovers $1 - \alpha$ in default, while the junior bond is fully wiped out. Therefore, the benchmark break-even rates on the senior loan and junior bond are

$$y_l^* = \frac{(\alpha - F_b)\rho_0}{(1 - F_b)(1 - \rho_0)}, \quad y_b^* = \frac{\rho_0}{1 - \rho_0}. \quad (6)$$

The bond spread is therefore higher than the loan spread purely because the bond is junior. Since $LGD_b = 1$ and $LGD_l = \frac{\alpha - F_b}{1 - F_b}$, the seniority-adjusted bond rate is⁵

$$y_{b,adj}^* \equiv y_b^* \frac{LGD_l}{LGD_b} = \frac{(\alpha - F_b)\rho_0}{(1 - F_b)(1 - \rho_0)} = y_l^*. \quad (7)$$

Thus, absent renegotiation, the loan premium is zero after adjusting for seniority. Appendix A.2 derives these benchmark rates and shows how the seniority adjustment removes the mechanical spread difference between loans and bonds.

⁵Because default is a firm-level event, the probability of default is the same for both instruments, and differences in spreads reflect differences in loss given default. This allows us to adjust the bond rate for seniority by scaling with the ratio of LGDs.

Relative to the pure bank-debt case, the key new feature is that renegotiation now requires coordination with dispersed bondholders. To keep the model tractable, I capture this friction in reduced form by assuming that when the bank attempts to renegotiate in distress, restructuring fails with probability h and succeeds with probability $1-h$, where a higher h corresponds to more severe holdout frictions. Appendix A.2 derives the implied pricing and feasibility conditions under this setup. Intuitively, holdout risk should be more severe when the firm relies more heavily on bond financing, because restructuring then requires coordination among a larger pool of dispersed creditors.

If renegotiation succeeds, the firm continues to date 2 and defaults there with probability ρ_1^L . If renegotiation fails, the firm is liquidated immediately. In this setting, the break-even loan and bond rates are

$$y_l^h = \frac{(\alpha - F_b)\rho_0[h + (1-h)\rho_1^L] + \rho_0 C}{(1 - F_b)(1 - \rho_0)}, \quad (8)$$

and

$$y_b^h = \frac{\rho_0[h + (1-h)\rho_1^L]}{1 - \rho_0}. \quad (9)$$

The corresponding seniority-adjusted bond rate is

$$y_{b,adj}^h \equiv y_b^h \frac{LGD_l}{LGD_b} = \frac{(\alpha - F_b)\rho_0[h + (1-h)\rho_1^L]}{(1 - F_b)(1 - \rho_0)}. \quad (10)$$

After adjusting the bond rate for seniority, the loan premium becomes

$$\pi_L^h = y_l^h - y_{b,adj}^h = \frac{\rho_0 C}{(1 - F_b)(1 - \rho_0)}. \quad (11)$$

The key point is that h drops out of the conditional premium. Conditional on renegotiation,

holdout risk affects loan and bond payoffs symmetrically through the timing of default. After adjusting for seniority, this common component cancels out, leaving only compensation for the bank's renegotiation cost. Holdout risk therefore matters mainly through whether renegotiation remains feasible.

Appendix A.2 also derives the renegotiation-feasibility condition in the mixed-borrowing case and shows how holdout frictions shrink the renegotiation region. Shareholders prefer renegotiation only if

$$C < \bar{C}_h, \quad \bar{C}_h = (1 - h)(1 - \rho_1^L)(\alpha + V_2 - 1) = (1 - h)\bar{C}. \quad (12)$$

Thus, holdout risk affects the loan premium mainly through feasibility: a rise in h shrinks the set of firms for which renegotiation is viable and can eliminate the premium altogether. The equilibrium loan premium is therefore

$$\pi_L = \begin{cases} \frac{\rho_0 C}{(1 - F_b)(1 - \rho_0)}, & \text{if } C < \bar{C}_h, \\ 0, & \text{if } C \geq \bar{C}_h. \end{cases} \quad (13)$$

To capture the idea that bond-intensive firms are more exposed to creditor coordination problems, I let holdout risk depend on the bond share, $h = h(F_b)$, with $h'(F_b) > 0$.

Proposition 2. *In the mixed borrowing case, the seniority-adjusted loan premium has the following properties:*

- **No renegotiation:** *Absent renegotiation, the seniority-adjusted bond rate equals the loan rate, so the loan premium is zero.*

- **Conditional premium:** *If renegotiation takes place,*

$$\pi_L^h = \frac{\rho_0 C}{(1 - F_b)(1 - \rho_0)},$$

so the premium increases in the renegotiation cost C and in the bond share F_b .

- **Feasibility condition:** *Renegotiation is feasible only if*

$$C < \bar{C}_h = (1 - h)\bar{C}.$$

A rise in holdout risk reduces the range of firms for which renegotiation is feasible and can eliminate the premium altogether.

- **Cross-sectional implication:** *If holdout risk depends on bond reliance, $h = h(F_b)$ with $h'(F_b) > 0$, then an increase in holdout frictions reduces the equilibrium loan premium more strongly for bond-intensive firms.*

This proposition motivates the empirical design. In the data, I measure the loan premium as the difference between the observed loan spread and the bond-implied loan spread, which is the empirical counterpart of the model-implied premium π_L . A legal change that makes out-of-court restructuring harder can be represented in the model as an increase in holdout risk h . Equation (13) implies that such a change reduces the value of renegotiation flexibility and can eliminate the loan premium for firms near the renegotiation threshold. If holdout risk rises more strongly with bond reliance, the decline in the loan premium is larger for bond-intensive firms. Empirically, I proxy for bond reliance using measures of firms' bond intensity, which correspond to the model parameter F_b , and I use the Marblegate ruling as an exogenous increase in restructuring frictions. The empirical analysis therefore tests a reduced-form implication of

the model: when out-of-court restructuring becomes less feasible, the premium firms pay for bank renegotiation flexibility should fall, especially for firms that rely more heavily on bond financing.

2.3 The Value of the Prepayment Option

Bank loans also embed prepayment flexibility. When a firm’s credit quality improves, the borrower may refinance or renegotiate the loan at a lower spread (Roberts and Sufi, 2009; Roberts, 2015). This option is valuable to borrowers but costly to lenders because it truncates future upside. I model this force in a simple two-period setting. Appendix A.4 derives the prepayment premium and shows how it depends on the likelihood and magnitude of future credit improvement.

The model delivers two implications. First, if prepayment risk is priced into the loan spread, the required compensation is positive but modest. Under plausible calibrations, it amounts to roughly 20–30 basis points, much smaller than the component associated with distressed renegotiation. Second, the prepayment component is increasing in the likelihood and magnitude of future credit improvement, because lenders are more exposed to repricing when borrowers are more likely to refinance into cheaper contracts.

These implications motivate the empirical tests in Section 5. If the loan premium partly compensates for prepayment risk, then higher loan premia should forecast subsequent rating improvements. But because the magnitude is small, prepayment flexibility should explain only a limited share of the overall loan premium.

3 Data

This section describes the data sources and sample construction for the empirical analysis. I start from syndicated loan originations in DealScan and match them to secondary-market corporate bond quotes from Refinitiv to measure the loan premium relative to public debt. I then combine these data with firm financial information from Compustat and S&P Capital IQ and stock return data from CRSP. These datasets provide the inputs needed to construct the loan premium and the firm-level controls used in the analysis.

Loan origination: I retrieve syndicated loan origination data from DealScan for the period 1997 to 2025, which contains historical information on loan pricing, contract terms, and other facility characteristics. Before matching the loans to company information, I apply the following filters: 1) the borrower is a U.S. firm and the loan tranche currency is USD; 2) I exclude loans to financial firms (SIC 6000-6999) and quasi-public firms (SIC above 8999); 3) the loan type is ‘Term Loan’, ‘Revolver/Term Loan’, or ‘Revolver > 1Y’; 4) I exclude sponsored loans and loans with the purpose of ‘Commercial paper backup’, ‘Debtor-in-possession’, ‘Exit financing’, ‘Leveraged Buyout’, ‘Management Buyout’, or ‘Sponsored Buyout’; 5) the loan is senior; and 6) the loan’s margin interest rate is relative to LIBOR or SOFR. I then use the linking table provided by [Chava and Roberts \(2008\)](#) to map DealScan borrowers to Compustat and supplement this match by identifying issuers’ permanent IDs in Refinitiv and retrieving their CUSIPs.

Firm characteristics: Firm characteristics are primarily obtained from Compustat and Capital IQ. From Compustat quarterly data, I retrieve book assets, total liabilities, book equity, total long-term debt, total revenue, operating income, and net property, plant, and equipment. To construct the structural-model-based measure of the loan premium, I also retrieve detailed

capital structure information from Capital IQ, including total bank debt, the outstanding balance of capital leases, and total undrawn credit.

Bond issues and bond prices: From Mergent FISD, I collect bonds issued by firms in the loan sample. After merging bond issues with loan originations, I apply the following filters: 1) the bond is issued by a U.S. firm and denominated in USD; 2) the bond is not perpetual, and the maturity difference between the bond and the loan does not exceed two years; and 3) the bond is issued before the loan origination. I then obtain daily yield-to-maturity quotes for the matched bonds from Refinitiv. For each loan, I take the average bond yield over the 10 calendar days preceding the loan origination and merge these quotes back into the loan dataset. If multiple bonds are matched to the same loan, I retain the bond with maturity closest to that of the loan.

Bond holdings: To test the renegotiation channel, I measure each firm’s outstanding bond holdings at the time of loan origination. I again use Mergent FISD to collect bond-level information, but unlike the matched-bond sample, I do not impose a maturity restriction. After matching bond issues to loan originations, I apply only two filters: 1) the bond is issued by a U.S. firm and denominated in USD; and 2) the bond is issued before the loan origination. I then use bond status information and status effective dates from Refinitiv to identify whether each matched bond remains active with a positive amount outstanding. Bonds that have matured or have been called with zero outstanding amount are excluded. For bonds issued by the same firm on the same date, I also compare issue amount, coupon rate, and other characteristics and remove duplicates that likely reflect the same security issued under different formats, such as Rule 144A or private placements.

Other data sources: I obtain the LIBOR/SOFR swap curve from Bloomberg to construct maturity-matched risk-free rates. Stock price and return data from CRSP are used to compute

market capitalization, stock returns, and equity volatility. Historical S&P long-term issuer ratings are collected from Capital IQ and translated into numerical scores as shown in Appendix Table A.I.

3.1 Sample Construction

I define the loan premium as the difference between the actual loan spread and the bond-implied loan spread at the time of loan origination. In the theoretical framework, the relevant comparison is between the observed loan spread and a seniority-adjusted benchmark for comparable public debt. Empirically, I operationalize this benchmark using a recovery-adjusted structural model of credit risk that builds on [Merton \(1974\)](#) and is implemented following [Schwert \(2020\)](#). This is the main measure used in the regression analysis because it allows the implied spread on senior bank debt to vary with the firm’s capital structure, debt seniority, and asset risk, while preserving the seniority-adjusted comparison emphasized by the model.

For comparability with prior work, I also construct a reduced-form measure following [Duffie and Singleton \(1999\)](#) and [Schwert \(2020\)](#). Under the assumption that cross-default provisions equalize the probability of default across bonds and loans issued by the same firm on the same date, differences in spreads reflect differences in loss given default, so the matched bond spread can be scaled by the relative LGD of loans and bonds. Following [Schwert \(2020\)](#), I set $LGD_{loan}/LGD_{bond} = 1/4$. This reduced-form measure is used only as a benchmark and not in the main regression analysis.

I now describe the construction of the structural-model-based bond-implied loan spread. Assume the firm’s asset value follows a geometric Brownian motion,

$$d \ln V_t = \left(r - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t^Q. \quad (14)$$

The firm has two classes of zero-coupon debt: senior loan debt with face value K_S and junior bond debt with face value K_J , both maturing at time T . Following [Glover \(2016\)](#), I assume that a fraction α of firm value is lost in default, where α is proxied by $(0.45 - 0.2Lev_{book})$ and Lev_{book} denotes book leverage. The value of the senior debt, denoted as D_S , is given by:

$$D_S = (1 - \alpha)(1 - \Phi(d_{1,S}))V + K_S e^{-rT} \Phi(d_{2,S}),$$

$$\text{with } d_{1,S} = \frac{\ln\left(\frac{V}{\min(K_S/(1-\alpha), K_S+K_J)}\right) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_{2,S} = d_{1,S} - \sigma\sqrt{T}. \quad (15)$$

The value of the junior debt, denoted as D_J , is given by:

$$D_J = (1 - \alpha) \left[V(\Phi(d_{1,S}) - \Phi(d_1)) - K_S e^{-rT} (\Phi(d_{2,S}) - \Phi(d_2)) \right] + K_J e^{-rT} \Phi(d_2),$$

$$\text{with } d_1 = \frac{\ln(\frac{V}{K_S+K_J}) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (16)$$

The yields of senior and junior debt, denoted as y_S and y_J , are

$$y_S = \frac{1}{T} \ln \frac{K_S}{D_S}, \quad \text{and } y_J = \frac{1}{T} \ln \frac{K_J}{D_J}. \quad (17)$$

I implement the model in four steps:

- First, I recover the junior debt price D_J from the observed bond yield y_J using Equation 17.
- Second, I use K_S , K_J , V , T , r , and D_J in Equation 16 to back out the asset volatility σ .
- Third, I substitute these parameters into Equation 15 to obtain the price of senior debt D_S .
- Finally, I compute the implied yield on senior debt, which is the bond-implied loan yield

y_S .

To implement the structural model, I use detailed debt structure information from Capital IQ. I define the senior debt amount K_S as the sum of bank debt, capital leases, and undrawn credit, and define junior debt K_J as total book debt minus senior debt. Detailed definitions and sources for all model inputs are reported in Table 1.

Table 1: Structural-model inputs used to compute the bond-implied loan spread. This table reports the parameters used in the recovery-adjusted structural model, their economic interpretation, the empirical measures used to proxy them, and the underlying data sources.

Parameter	Description	Data	Source
K	Total book debt	Long-term debt plus current liabilities	Compustat
K_S	Senior debt amount	Bank, lease and undrawn debt	Capital IQ
K_J	Junior debt amount	Total book debt minus senior debt	Compustat
V	Quasi-market asset	Total book debt plus market equity	Compustat, CRSP
T	Maturity	Loan and bond maturities	DealScan, FISD
r	Risk-free rate	Maturity-matched LIBOR/SOFR swap rate	Bloomberg
y_J	Junior debt yield	Daily quote of bond's yield	Refinitiv

3.2 Summary Statistics

This subsection summarizes the main variables used in the empirical analysis. The variable of interest is the loan premium, defined as the actual loan spread minus the bond-implied loan spread constructed from the structural model. Table 2 reports summary statistics for the loan premium and the key firm and debt characteristics in the full sample and in the term-loan-only sample. Detailed definitions of the variables are provided in Appendix Table A.II.

Panel (A) of Table 2 presents the summary statistics of the full sample, consisting of both term loans and credit lines over one year. On average, the loan spread is around 191 bps, and the initial time to maturity of the loans is 4.4 years.⁶ For the matched bond observations,

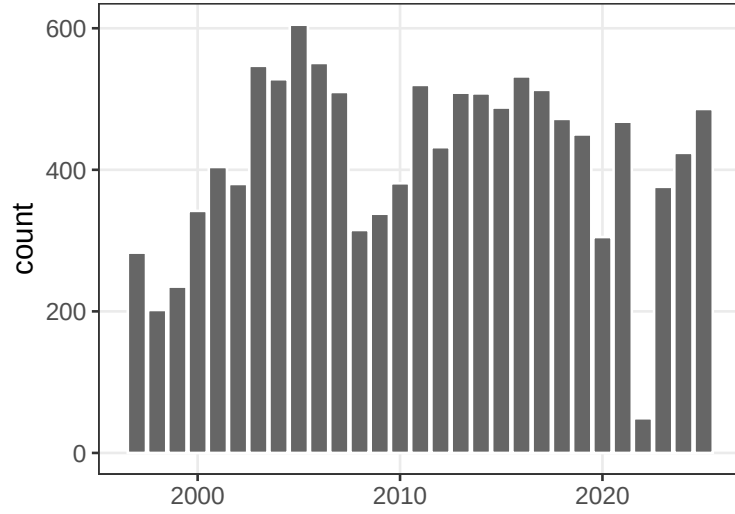
⁶By definition, the all-in-drawn spread is calculated as the sum of the interest rate margin and various fees. However, as also documented by [Berg et al. \(2016\)](#), I find that the reported all-in-drawn spread is extremely close to the interest rate margin. Thus, in this paper, I treat the all-in-drawn spread as the rate that the borrower pays to all syndicate members in a loan contract.

Table 2: Summary statistics. Panel A reports summary statistics for the full sample, and Panel B reports summary statistics for the term-loan-only sample. The loan premium is defined as the actual loan spread minus the bond-implied loan spread from the structural model. The loan premium series are winsorized at the 0.1% level.

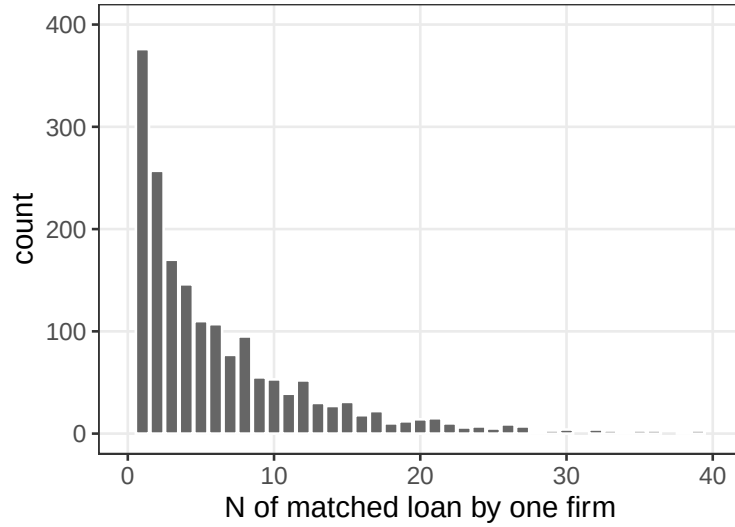
Panel A: Full sample								
Variable	N	Mean	SD	Min	P25	Median	P75	Max
<i>Loan characteristics</i>								
All-in-drawn spread (bps)	12,153	191	121	5	112	162	250	1,200
Tranche amount (\$M)	12,153	863	1,177	0	200	500	1,000	24,000
Maturity	12,153	4.4	1.3	1	3.4	5	5	12
Term loan	12,153	0.28	0.45	0	0	0	1	1
Secured loan	12,153	0.47	0.5	0	0	0	1	1
Lead arranger count	12,153	3.5	3.3	0	1	2	5	29
Performance pricing	12,153	0.29	0.45	0	0	0	1	1
Maturity matched rf (%)	12,153	3.1	1.8	0.15	1.6	2.8	4.4	7.6
<i>Bond characteristics</i>								
Yield to maturity (%)	12,153	6.8	5.5	0.026	3.8	5.6	7.9	50
Yield spread (bps)	12,153	372	532	-558	89	216	447	4,841
Face value (\$M)	12,153	433	420	0.001	200	315	500	9,000
Maturity	12,153	4.5	1.5	0.0082	3.5	4.6	5.5	13
Maturity matched rf (%)	12,153	3.1	1.8	0.18	1.5	2.9	4.4	7.6
<i>Firm characteristics</i>								
Market asset (\$B)	9,688	24	62	0.064	2.4	6.9	20	1,490
Market leverage (%)	9,688	41	22	0	23	38	56	98
Asset volatility	9,540	0.22	0.12	0.013	0.14	0.2	0.27	1.4
Distance to default	9,537	4.9	10	-398	3.2	5.1	7.3	54
Trailing stock return (%)	9,585	18	179	-96	-19	6	29	11,730
Asset market to book (%)	9,688	129	102	9.5	79	101	143	2,783
Asset tangibility (%)	11,866	39	27	0	15	36	63	98
Profitability (%)	11,628	2.9	2.8	-51	1.8	2.8	4	36
Bond to book asset (%)	11,982	32	29	0.26	16	25	39	621
S&P rating score	9,295	11	3.3	1	9	12	13	25
<i>Loan premium</i>								
Reduced form (bps)	12,153	98	123	-827	55	96	149	662
Structural form (bps)	4,910	59	270	-2,742	16	96	169	2,272

Table 2 Continued

Panel B: Term loan only								
Variable	N	Mean	SD	Min	P25	Median	P75	Max
<i>Loan characteristics</i>								
All-in-drawn spread (bps)	3,350	263	146	5	175	250	325	1,200
Tranche amount (\$M)	3,350	563	655	0	150	340	743	5,630
Maturity	3,350	4.8	1.7	1	3.5	5	6	12
Term loan	3,350	1	0	1	1	1	1	1
Secured loan	3,350	0.72	0.45	0	0	1	1	1
Lead arranger count	3,350	3.3	3.2	0	1	2	5	19
Performance pricing	3,350	0.17	0.38	0	0	0	0	1
Maturity matched rf (%)	3,350	3.2	1.9	0.18	1.6	2.9	4.8	7.6
<i>Bond characteristics</i>								
Yield to maturity (%)	3,350	8.1	6.2	0.026	4.8	6.6	9.3	50
Yield spread (bps)	3,350	481	591	-558	177	317	549	4,805
Face value (\$M)	3,350	454	387	0.001	200	350	546	4,800
Maturity	3,350	5	1.9	0.0082	3.7	5	6.2	13
Maturity matched rf (%)	3,350	3.3	1.9	0.18	1.7	3	4.8	7.6
<i>Firm characteristics</i>								
Market asset (\$B)	2,611	15	47	0.064	2.2	5.7	14	722
Market leverage (%)	2,611	49	22	0.23	32	47	66	98
Asset volatility	2,541	0.21	0.12	0.013	0.13	0.19	0.26	1.4
Distance to default	2,540	3.6	15	-398	2.5	4.3	6.3	54
Trailing stock return (%)	2,559	20	144	-96	-25	3.7	31	3,701
Asset market to book (%)	2,611	124	109	9.5	78	100	136	2,783
Asset tangibility (%)	3,243	36	25	0	14	34	55	97
Profitability (%)	3,170	2.8	2.8	-50	1.8	2.8	3.9	29
Bond to book asset (%)	3,267	36	33	0.26	17	27	44	524
S&P rating score	2,538	13	2.8	3	12	13	14	25
<i>Loan premium</i>								
Reduced form (bps)	3,350	143	140	-765	87	142	204	662
Structural form (bps)	1,745	93	293	-2,742	40	122	200	2,272



(a) Loan issuance



(b) Loan issuance among issuers

Figure 3: Distribution of loan originations in the matched sample. Panel (a) shows the time-series distribution of loan originations in the final sample. Loan issuance is relatively evenly distributed across most years, but declines sharply during market downturns, particularly in 2008, 2020, and 2022. Panel (b) shows the distribution of loan originations across issuers. It reports how often each borrower appears in the final sample. For example, the first bar indicates that approximately 370 borrowers appear only once in the matched loan-bond sample.

the spread over the maturity-matched swap rate is close to 372 bps. Due to the maturity restriction, the average time to maturity of the bonds is 4.5 years. The average size of the

bonds' face value is roughly half of the loan size. Regarding firm characteristics, the median market asset is around \$6.9 billion, indicating a sample of relatively large firms. The typical issuer rating is *BB*, near the boundary between investment grade and non-investment grade. On average, the sample includes firms with a positive profitability of 2.9%. The statistics of the structural loan premium are presented at the bottom of Table 2. The median structural loan premium is close to 96 bps, indicating a significant spread between the private loan market and the public bond market.

Panel (B) of Table 2 presents the summary statistics when the sample is restricted to term loans only. The median structural loan premium in this sample rises to 122 bps, aligning more closely with the estimates reported by [Schwert \(2020\)](#), who examines term loans exclusively. Both the all-in-drawn spread and the matched bond spread are larger in the term-loan-only sample than in the full sample. The term-loan-only sample remains broadly comparable to the full sample, although firms in this sample are somewhat smaller, more levered, and lower rated on average.

I also present the histogram of loan originations in Figure 3. Panel (a) provides an overview of the sample distribution over time. Loan issuance is relatively evenly distributed across most years, but there are sharp declines during market downturns, particularly in 2008, 2020, and 2022. Panel (b) illustrates the frequency of each borrower's appearance in the final sample. For example, the first bar indicates that approximately 370 borrowers have only one loan origination with matched bond quotes in the final sample.

Figure 4 plots the distribution of bond holdings scaled by book assets. The distribution is right-skewed but exhibits substantial cross-sectional variation, supporting the use of bond holdings as a meaningful measure of firms' reliance on public debt in the subsequent analysis.

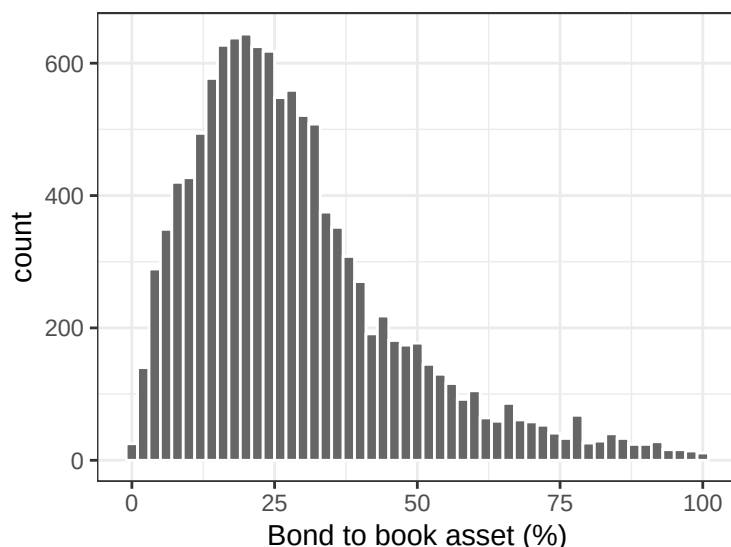


Figure 4: Distribution of bond holdings scaled by book assets. This figure plots the distribution of bond holdings scaled by book assets in the matched loan sample. The distribution is right-skewed, with substantial cross-sectional variation in reliance on bond financing.

4 Testing the Distressed Renegotiation

This section tests whether the loan premium compensates banks for providing renegotiation flexibility in financial distress. I use the Marblegate ruling as a shock to the feasibility of out-of-court restructuring and examine how the loan premium responds across firms with different reliance on bond financing. I begin by describing the institutional setting and why the ruling mattered for distressed bond restructurings.

4.1 Institutional Background of Marblegate

The TIA of 1939 was enacted to provide protections to holders of debt securities, with Section 316(b) stating:

"the right of any holder of any indenture security to receive payment of the principal of and interest on such indenture security, on or after the respective due dates

expressed in such indenture security, or to institute suit for the enforcement of any such payment on or after such respective dates, shall not be impaired or affected without the consent of such holder..."

In 2014, the for-profit education company Education Management Corp. ("EDMC") implemented an out-of-court exchange offer that restructured approximately \$1.3 billion in secured debt and \$217 million in unsecured notes issued by its subsidiaries. Although the transaction did not formally amend the payment terms of the unsecured notes, dissenting noteholders were effectively left with claims against entities that no longer held meaningful assets. As a result, investors who declined to participate in the exchange offer would have little practical ability to recover on their claims.

Two hold-out noteholders, collectively referred to as "Marblegate," sued in October 2014, arguing that the exchange offer violated Section 316(b) by depriving them of the practical ability to receive payment and by coercing participation. Although the court denied Marblegate's request for an injunction, it ruled that the plaintiffs were likely to succeed on the merits of their TIA claim. This ruling immediately raised concerns that minority bondholders could gain greater leverage to block distressed exchange offers even when the formal payment terms of the bonds were left unchanged.

The main significance of the ruling was its reinterpretation of the TIA as offering broad protection against nonconsensual restructurings, rather than merely prohibiting majority amendment of a limited set of core payment terms. If adopted more broadly, this interpretation would make out-of-court restructurings more difficult to complete without unanimous consent, even when the underlying actions were otherwise permitted by the indenture. More generally, the ruling created uncertainty around the feasibility of distressed exchange offers and increased concern that stronger holdout incentives could push more firms into formal bankruptcy.

The overturn of Marblegate in 2017: The defendants filed for a review in the Second Circuit Court of Appeals, where the appeals court largely overturned Marblegate. The Second Circuit found the meaning of Section 316(b) ambiguous and resorted to the legislative history to determine its meaning. In January 2017, the Second Circuit decision restored the law to its pre-Marblegate interpretation. Among other things, the decision provided bond issuers with comfort that they could again seek to implement exchange offers through the use of exit consents without the uncertainty of violating Section 316(b).

4.2 Empirical Strategy and Baseline Results

The empirical objective is to test whether the loan premium compensates banks for providing renegotiation flexibility in financial distress. The model predicts that an increase in holdout frictions reduces the value of this flexibility and lowers the loan premium, with a larger decline for firms that rely more heavily on bond financing.

I interpret the Marblegate ruling as a shock of this type. In the empirical analysis, the loan premium is measured as the difference between the observed loan spread and the bond-implied loan spread, while firms' bond reliance is the empirical counterpart of the model parameter F_b . The Marblegate ruling therefore captures an increase in restructuring frictions, corresponding to a rise in holdout risk, h . I estimate a difference-in-differences specification that compares the change in the loan premium after Marblegate between firms with high and low reliance on bond financing. The coefficient of interest is the interaction between the post-Marblegate indicator and bond reliance, which the model predicts should be negative.

A standard difference-in-differences regression compares treated and untreated groups before and after a shock. However, the Marblegate ruling was reversed in 2017. Because the effects of the reversal may be attenuated by anticipation or gradual adjustment, I first restrict the analysis

to the period before 2017 (the *restricted sample*) and estimate the following specification:

$$\text{Loan premium}_{i,t} = \beta_1 M_t + \beta_2 B_{i,t} + \beta_3 (M_t \times B_{i,t}) + \mathbf{x}_{i,t} \boldsymbol{\gamma} + \tau_s + e_{i,t}, \quad (18)$$

where M_t is an indicator equal to one for loans originated in 2015 or 2016. The variable $B_{i,t}$ is an indicator for bond-intensive firms. Following Kornejew (2025), I measure bond intensity as outstanding bond face value scaled by book assets and classify a firm as bond-intensive if this ratio exceeds the sample median of 0.26.⁷ This cutoff is very close to the median bond intensity of 0.25 reported by Kornejew (2025).

The key variable of interest is the interaction between the bond-intensive indicator and the Marblegate indicator ($M_t \times B_{i,t}$), and the coefficient $\hat{\beta}_3$ captures whether the post-Marblegate change in the loan premium is larger for bond-intensive firms. The specification in Equation 18 also includes the industry fixed effect, as captured by τ_s . I do not include the firm fixed effect since it would largely remove the statistical power: firms with only one loan observation would be dropped, and firms with only a few loan observations would offer very limited within-firm variation (see Figure 3 for the sample distribution). Nevertheless, I include a set of firm-level control variables: market leverage, asset volatility, trailing stock return, asset market-to-book ratio, asset tangibility, and profitability, which are collectively denoted by $\mathbf{x}_{i,t}$. I cluster the standard errors at the firm and year levels. The first two columns in Table 3 summarize the test results.

The coefficient of interest is the interaction $M_t \times B_{i,t}$. Its estimate is negative and statistically significant across specifications, consistent with the model prediction that Marblegate should reduce the loan premium more for firms with greater exposure to bondholder holdout frictions.

⁷Using book debt instead of book assets, or using a time-varying cutoff instead of the full-sample median, leads to very similar conclusions.

Table 3: Loan premium and the Marblegate ruling. This table reports difference-in-differences estimates of the response of the loan premium to the Marblegate ruling, using the sample of loans originated through 2016. The dependent variable is the loan premium in basis points, defined as the seniority-adjusted difference between the loan spread and the matched bond-implied loan spread. M_t is an indicator equal to one for loans originated in 2015 or 2016, and $B_{i,t}$ is an indicator equal to one if the issuer's bond holdings scaled by book assets exceed the sample median of 0.26. The coefficient of interest is the interaction $M_t \times B_{i,t}$. The first two columns estimate the baseline specification with industry fixed effects and firm-level controls. The last column adds year fixed effects, in which case M_t is absorbed and omitted. Standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>		
	Loan premium		
	(1)	(2)	(3)
Marblegate	15.77 (26.55)	36.32 (25.45)	
Bond intensive	60.15*** (11.94)	104.52*** (14.25)	91.47*** (15.92)
Marblegate*Bond intensive	−83.24*** (22.17)	−107.31*** (24.46)	−94.09*** (25.29)
Covariates	No	Yes	Yes
Industry FE	Yes	Yes	Yes
Year FE	No	No	Yes
Observations	3,202	3,010	3,010
Adjusted R ²	0.06	0.11	0.13
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01			

The estimated effect is economically large, ranging from 83 to 107 basis points.

The coefficient on $B_{i,t}$ is positive and significant, indicating that before the shock bond-intensive firms carried higher loan premia. By contrast, the coefficient on M_t is positive but statistically insignificant, implying no comparable post-Marblegate change for the omitted group of non-bond-intensive firms. This is consistent with the model, which predicts a differential decline for firms more exposed to restructuring frictions rather than a uniform decline across all firms.

I then re-estimate the specification on the *restricted sample* including year fixed effects, denoted by θ_t . Because the Marblegate indicator is absorbed by the year fixed effects, it is

omitted from the regression:

$$\text{Loan premium}_{i,t} = \beta_2 B_{i,t} + \beta_3 (M_t \times B_{i,t}) + \mathbf{x}_{i,t} \boldsymbol{\gamma} + \tau_s + \theta_t + e_{i,t}. \quad (19)$$

The coefficient of interest remains the interaction term $M_t \times B_{i,t}$. As shown in the last column of Table 3, its estimate remains negative and statistically significant, with a magnitude close to the baseline specification. This suggests that the differential post-Marblegate decline in the loan premium for bond-intensive firms is unlikely to be driven by common year-specific factors.

4.3 Heterogeneity in Renegotiation Frictions

The baseline DID estimates establish that the loan premium declines more for bond-intensive firms after Marblegate. I next examine whether this effect is stronger in settings where renegotiation is likely to be easier ex ante and weaker where renegotiation is already more difficult.

I first consider whether the effect varies with institutional loan ownership. As documented by Demiroglu and James (2015), loans held by institutional investors are harder to restructure out of court than loans held by traditional bank lenders. Because DealScan does not directly identify CLO ownership, I proxy institutional ownership using the presence of a secondary-market identifier (LIN code) and define $LIN_{i,t} = \mathbb{1}(\text{LIN available})$. I then estimate

$$\text{Loan premium}_{i,t} = \beta_3 (M_t \times B_{i,t}) + \delta (M_t \times B_{i,t} \times LIN_{i,t}) + \mathbf{x}_{i,t} \boldsymbol{\gamma} + \mathbf{z}_{i,t} \boldsymbol{\phi} + \tau_s + e_{i,t}. \quad (20)$$

The triple interaction term captures whether the Marblegate effect differs between LIN-available and LIN-unavailable loans. The vector $\mathbf{z}_{i,t}$ includes the associated lower-order terms. Table 4 reports the results for the *restricted sample*.

Table 4: Heterogeneity in the Marblegate effect by institutional loan ownership. This table reports triple-difference estimates of the response of the loan premium to the Marblegate ruling in the sample of loans originated through 2016. The dependent variable is the loan premium in basis points. M_t is an indicator equal to one for loans originated in 2015 or 2016, $B_{i,t}$ is an indicator equal to one if the issuer’s bond holdings scaled by book assets exceed the sample median of 0.26, and $LIN_{i,t}$ is an indicator equal to one if the loan has a secondary-market identifier, which serves as a proxy for institutional ownership or CLO association. The coefficient of interest is the triple interaction $M_t \times B_{i,t} \times LIN_{i,t}$. The associated lower-order terms are included in the regressions but omitted from the table. Standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>	
	Loan premium	
	(1)	(2)
Marblegate*Bond intensive	−129.15*** (13.28)	−162.53*** (18.07)
Marblegate*Bond intensive*LIN indicator	109.72** (47.59)	137.40*** (41.99)
Covariates	No	Yes
Industry FE	Yes	Yes
Year FE	No	No
Observations	3,202	3,010
Adjusted R ²	0.06	0.11
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Across all specifications, the coefficients on the baseline interaction $M_t \times B_{i,t}$ remain negative, while the coefficients on the triple interaction $M_t \times B_{i,t} \times LIN_{i,t}$ are positive. This pattern implies that the Marblegate effect on the loan premium is less negative for CLO-associated loans than for non-CLO-associated loans. In other words, although bond-intensive firms generally experience a decline in the loan premium after the shock, the decline is smaller for loans associated with institutional investors. This finding is consistent with the view that such loans were already harder to renegotiate before Marblegate and therefore experienced a more limited additional response to the ruling.

I next consider whether the treatment effect varies with the number of parties involved in renegotiation. Loans with fewer key lenders should be easier to restructure out of court and

Table 5: Heterogeneity in the Marblegate effect by syndicate structure. This table reports triple-difference estimates of the response of the loan premium to the Marblegate ruling in the sample of loans originated through 2016. The dependent variable is the loan premium in basis points. M_t is an indicator equal to one for loans originated in 2015 or 2016, $B_{i,t}$ is an indicator equal to one if the issuer’s bond holdings scaled by book assets exceed the sample median of 0.26, and $OneLead_{i,t}$ is an indicator equal to one if the loan has only one lead arranger. The coefficient of interest is the triple interaction $M_t \times B_{i,t} \times OneLead_{i,t}$. The associated lower-order terms are included in the regressions but omitted from the table. Standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>	
	Loan premium	
	(1)	(2)
Marblegate*Bond intensive	−51.87* (31.07)	−66.97* (34.35)
Marblegate*Bond intensive*One lead indicator	−181.08*** (62.50)	−205.99*** (67.42)
Covariates	No	Yes
Industry FE	Yes	Yes
Year FE	No	No
Observations	3,201	3,009
Adjusted R ²	0.06	0.11
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

therefore more exposed to a shock that raises holdout frictions.

Because most loans in the sample are syndicated, I focus on the number of lead arrangers rather than the total number of lenders. I define $OneLead_{i,t} = \mathbb{1}(\text{number of lead arrangers} = 1)$ and estimate

$$\text{Loan premium}_{i,t} = \beta_3(M_t \times B_{i,t}) + \delta(M_t \times B_{i,t} \times OneLead_{i,t}) + \mathbf{x}_{i,t}\boldsymbol{\gamma} + \mathbf{z}_{i,t}\boldsymbol{\phi} + \tau_s + e_{i,t}. \quad (21)$$

The results for the *restricted sample* are summarized in Table 5. The coefficient on $M_t \times B_{i,t}$ remains negative, indicating that among loans with multiple lead arrangers, bond-intensive firms experience a larger decline in the loan premium after Marblegate. The coefficient on the triple interaction $M_t \times B_{i,t} \times OneLead_{i,t}$ is also negative, implying an additional decline for

loans with only one lead arranger. This pattern indicates that the Marblegate effect is strongest where renegotiation is likely to be easiest ex ante, namely when fewer parties are involved.

Taken together, these triple-difference results strengthen the interpretation of the baseline DID estimates. The Marblegate effect is weaker for loans associated with institutional investors, where renegotiation is already more difficult, and stronger for loans with only one lead arranger, where renegotiation is likely to be easier. This pattern is consistent with the view that the decline in the loan premium reflects a reduction in the value of bank renegotiation flexibility.

4.4 Dynamic Effects of Marblegate

The baseline results focus on the period after the 2014 Marblegate ruling and before its reversal in 2017. I next examine the timing and persistence of the effect using a dynamic difference-in-differences specification. Specifically, I interact the bond-intensive indicator $B_{i,t}$ with a set of event-time indicators relative to 2014, the last pre-ruling year:

$$\text{Loan premium}_{i,t} = \beta_2 B_{i,t} + (\mathbf{1}_{\pm T} \times B_{i,t}) \boldsymbol{\delta}_{\pm T} + \mathbf{x}_{i,t} \boldsymbol{\gamma} + \tau_s + \theta_t + e_{i,t}. \quad (22)$$

The coefficients on $\boldsymbol{\delta}_{\pm T}$ trace the differential evolution of the loan premium for bond-intensive firms relative to non-bond-intensive firms, with 2014 as the omitted benchmark year.

Figure 5 plots the estimated interaction coefficients. The estimates are close to zero in the pre-ruling years, which is consistent with the parallel-trends assumption and helps rule out differential pre-existing trends in loan premia between bond-intensive and non-bond-intensive firms. After Marblegate, the coefficients turn sharply negative in 2015 and 2016, indicating a sizable decline in the loan premium for bond-intensive firms. The estimate for 2017 remains negative, suggesting that the effect did not disappear immediately after the reversal and grad-

ually faded from 2018 onward. Overall, the event-study evidence is consistent with the view that Marblegate reduced the value of bank renegotiation flexibility by making out-of-court restructuring more difficult for firms with greater reliance on bond financing.

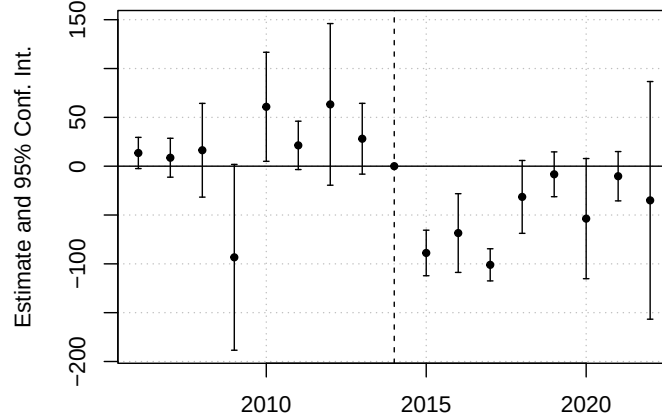


Figure 5: Dynamic effects of Marblegate on the loan premium. This figure plots the event-time interaction coefficients from the dynamic difference-in-differences specification, where the loan premium is measured using the structural model. The variable $B_{i,t}$ equals one if the issuer firm’s bond holdings scaled by book assets are above the sample median. Event time is centered at 2014, the last pre-ruling year, so the plotted coefficients show the differential evolution of the loan premium for bond-intensive firms relative to non-bond-intensive firms.

4.5 Interpreting the Decline in the Loan Premium

An important question is whether the decline in the loan premium after Marblegate reflects a reduction in the value of bank renegotiation flexibility, or instead a mechanical change in the pricing of the underlying debt instruments. To shed light on this issue, I examine whether the effect originates from the loan spread or the bond spread.

I first re-estimate the baseline difference-in-differences specification using the loan spread and the bond spread as alternative dependent variables. The results are reported in Table 6. When the dependent variable is the loan spread, the coefficient on the interaction between Marblegate and bond intensity is significantly negative, indicating that loan spreads decline for

bond-intensive firms after the shock. By contrast, using monthly bond spread or return from TRACE, I find no corresponding increase in bond spreads. These results indicate that the fall in the measured loan premium is driven primarily by a reduction in loan spreads rather than by a widening of bond spreads.

Table 6: Interpreting the Marblegate effect on the loan premium. This table re-estimates the baseline difference-in-differences specification using alternative dependent variables. Column (1) uses the loan spread in basis points. Columns (2) and (3) use monthly bond spread and monthly bond return data from TRACE, a transaction-based database of secondary-market corporate bond prices. The coefficient of interest is the interaction between the Marblegate indicator and the bond-intensive indicator. The loan-spread specification includes industry fixed effects, while the TRACE specifications include bond and month fixed effects. Standard errors are clustered at the issuer-firm and year levels in column (1), and at the bond and month levels in columns (2) and (3).

	<i>Dependent variable:</i>		
	Loan spread	Bond spread	Bond return
	(1)	(2)	(3)
Marblegate	1.93 (15.70)		
Bond intensive	63.49*** (6.89)		
Marblegate $\times$ Bond intensive	-40.43*** (9.08)	19.504 (25.122)	0.001 (0.001)
Covariates	No	No	No
Industry FE	Yes	No	No
Month FE	No	Yes	Yes
Bond FE	No	Yes	Yes
Observations	7,908	242,756	236,100
Adjusted R ²	0.13	0.554	0.042

Note:

*p<0.1; **p<0.05; ***p<0.01

A related concern is that Marblegate may have changed the relative recovery prospects of loans and bonds, thereby affecting the seniority adjustment used to construct the loan premium. To assess this possibility, I examine whether the effect differs by secured status, since a recovery-priority interpretation would be most plausible for secured loans. The corresponding estimates

are reported in Appendix Table A.IV. The estimates provide little evidence that the Marblegate effect differs systematically between secured and unsecured loans, which weakens a recovery-priority interpretation of the decline in the loan premium.

More broadly, the pattern of evidence is difficult to reconcile with a pure LGD-adjustment story. The decline in the loan premium is driven by lower loan spreads rather than higher bond spreads, and it is systematically related to proxies for renegotiation frictions such as CLO association and the number of lead arrangers. Taken together, these patterns are more consistent with a decline in the value of renegotiation flexibility than with a mechanical change in the seniority adjustment alone.

4.6 Additional Robustness Checks

I conduct three additional robustness checks. First, I re-estimate the baseline specification using the full-period sample, including observations after 2016, while continuing to define the Marblegate period as loan originations in 2015 and 2016. The estimated effect is only slightly smaller than in the restricted sample (Appendix Table A.V).

Second, I re-estimate the baseline specification in the term-loan-only sample, which provides a cleaner comparison to the standard loan-premium setting in [Schwert \(2020\)](#). The corresponding estimates are also reported in Appendix Table A.VI, and the main result remains present in that subsample.

Third, I address the possibility that the 2014–2016 oil crisis confounds the results. In Appendix Table A.VII, I first include an interaction between the Marblegate indicator and an oil-exposure indicator, and then exclude oil-exposed firms from the sample altogether. In both cases, the interaction between Marblegate and bond intensity remains significantly negative, indicating that the main result is not driven by oil-sector distress.

5 Testing the Prepayment Channel

In this section, I examine whether the loan premium partially reflects the prepayment and repricing flexibility embedded in bank loans. The model implies that compensation for prepayment risk should be larger when the borrower is more likely to experience future credit improvement, and when the expected improvement is larger. Because these expectations are not directly observable, I instead conduct two reverse regressions. First, I test whether the loan premium predicts subsequent credit improvement. Second, I test whether a higher loan premium predicts a larger improvement in credit quality. The first test is specified by,

$$I(\text{Rating improves}) = \beta_1 \text{Loan premium}_{i,t} + \beta_2 \text{Initial rating}_{i,t} + \mathbf{x}_{i,t}\boldsymbol{\gamma} + \tau_s + \theta_t + e_{i,t}. \quad (23)$$

The dependent variable is an indicator that equals one if the issuer’s rating improves within five years after the loan origination. If the rating remains unchanged or is downgraded, the variable is zero.⁸ The unconditional mean of the rating improvement indicator is roughly 30%. The key explanatory variable is the loan premium (in percentage points). The theoretical analysis suggests that if compensation for prepayment risk is part of the loan premium, a higher premium should predict future credit quality improvement. Thus, I expect the estimated coefficient of the loan premium $\hat{\beta}_1$ to be positive. The specification in Equation 23 also includes industry and year fixed effects, represented by τ_s and θ_t . Given that it is generally easier for lower-credit firms to have rating improvements and it is not possible for highest-credit firms to improve, I incorporate the firm’s credit rating before the loan issuance as a control variable. In the

⁸To determine whether the rating is upgraded, downgraded, or unchanged, I proceed as follows. First, I find the current credit rating of a firm when the loan is originated. Then, I identify the next assigned rating by S&P of the firm after the loan origination. If the assigned time exceeds five years after loan origination, I treat the rating as unchanged. Otherwise, if the assigned time is within five years, I take the newly assigned rating, which can be upgraded, downgraded, or unchanged.

largest model specification, I also include the same set of firm-level control variables as in the renegotiation test, denoted by $\mathbf{x}_{i,t}$. Standard errors are clustered at the firm and year levels.

Table 7: Loan premium and subsequent rating improvement. This table reports regressions of an indicator for whether the issuer’s credit rating improves within five years of loan origination on the loan premium in percentage points. All specifications include the issuer’s initial rating, industry fixed effects, and year fixed effects. Column (2) further includes firm-level controls. Standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>	
	I(Rating improves within 5 years)	
	(1)	(2)
Loan premium	0.011*** (0.004)	0.008** (0.003)
Initial rating	0.038*** (0.005)	0.062*** (0.007)
Covariates	No	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	3,469	3,307
Adjusted R ²	0.125	0.166
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

The results are summarized in Table 7. The coefficients on the loan premium are positive and statistically significant across specifications, indicating that a higher loan premium is associated with a greater likelihood of future credit improvement. In terms of economic magnitude, the coefficient in column (1) is 0.011, implying that a one-percentage-point increase in the loan premium predicts a 1.1 percentage point higher probability of a rating improvement. Given that the unconditional mean of credit improvement is 30%, this corresponds to roughly a 4% increase in the likelihood of an upgrade. Column (2) adds firm-level controls, and the positive relationship remains robust.

Next, I replace the dependent variable with the actual change in the credit rating score

within five years after the loan origination. The regression is specified as follows:

$$\text{Rating score change}_{i,t} = \beta_1 \text{Loan premium}_{i,t} + \beta_2 \text{Initial rating}_{i,t} + \mathbf{x}_{i,t} \boldsymbol{\gamma} + \tau_s + \theta_t + e_{i,t}. \quad (24)$$

As suggested by the model, a larger improvement in the borrower’s credit quality makes the prepayment option more valuable. In Equation 24, the dependent variable is the change in the numeric rating score over the five years following loan origination, where a negative number indicates a rating upgrade (see Table A.I). If a larger loan premium contains more compensation for prepayment risk, it should predict a larger upgrade, that is, a more negative score change. Therefore, I expect $\hat{\beta}_1$ to be negative in this specification.

The test results are presented in Table 8. The coefficients on the loan premium are negative across specifications, consistent with the prediction that higher premia are associated with larger subsequent rating upgrades. The coefficient in column (1) indicates that a one-percentage-point increase in the loan premium predicts a decline of 0.035 in the future rating score. Although the economic magnitude is modest, this result is in line with the model’s implication that prepayment compensation accounts for only a limited share of the overall loan premium.

As a robustness check, I augment the baseline regression with the firm’s 5-year CDS spread to address the concern that subsequent rating improvement may simply reflect high-yield credit. Table A.VIII shows that, after controlling for the initial credit rating, the CDS spread does not predict future rating-score changes. By contrast, the coefficient on the loan premium remains negative and statistically significant on the same sample, indicating that the predictive content of the loan premium is not subsumed by CDS spreads.

Table 8: Loan premium and subsequent rating-score change. This table reports regressions of the change in the issuer’s numeric credit rating score within five years of loan origination on the loan premium in percentage points. Lower values of the dependent variable indicate larger rating upgrades. All specifications include the issuer’s initial rating, industry fixed effects, and year fixed effects. Column (2) further includes firm-level controls. Standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>	
	Rating (score) change within 5 years	
	(1)	(2)
Loan premium	−0.035*** (0.012)	−0.025** (0.011)
Initial rating	−0.051*** (0.013)	−0.108*** (0.023)
Covariates	No	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	3,469	3,307
Adjusted R ²	0.091	0.133
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01		

6 Conclusion

This paper studies why firms pay more for bank loans than for comparable public bonds. I provide evidence that a substantial share of this loan premium compensates banks for providing renegotiation flexibility in financial distress. Using the Marbledgate ruling as a shock to the feasibility of out-of-court restructuring, I find that the premium falls sharply for bond-intensive firms after the ruling, consistent with a decline in the value of bank-facilitated renegotiation. The effect is stronger in settings where out-of-court renegotiation is easier ex ante. The decline is driven primarily by changes in loan spreads rather than bond spreads, supporting the interpretation that Marbledgate reduced the value of bank renegotiation flexibility rather than repriced seniority.

I also show that prepayment and repricing flexibility explain a smaller portion of the pre-

mium. Taken together, the results imply that bank debt is valued not only because it provides funding, but also because it embeds contractual flexibility that helps firms navigate both downside and upside states. More broadly, the findings suggest that creditor structure is an important determinant of debt pricing: when coordination frictions make out-of-court restructuring harder, the pricing advantage of bank renegotiation flexibility declines.

7 Bibliography

- Becker, B. and Ivashina, V. (2014). Cyclicalities of credit supply: Firm level evidence. *Journal of Monetary Economics*, 62(1):76–93.
- Berg, T., Saunders, A., and Steffen, S. (2016). The total cost of corporate borrowing in the loan market: Don’t ignore the fees. *Journal of Finance*, 71(3):1357–1392.
- Berger, A. N. and Udell, G. F. (1995). Relationship lending and lines of credit in small firm finance. *Journal of Business*, 68(3):351–381.
- Bharath, S. T., Dahiya, S., Saunders, A., and Srinivasan, A. (2011). Lending relationships and loan contract terms. *Review of Financial Studies*, 24(4):1141–1203.
- Brudney, V. (1992). Corporate bondholders and debtor opportunism: In bad times and good. *Harvard Law Review*, pages 1821–1878.
- Chava, S. and Roberts, M. R. (2008). How does financing impact investment? The role of debt covenants. *Journal of Finance*, 63(5):2085–2121.
- Chemmanur, T. J. and Fulghieri, P. (1994). Investment bank reputation, information production, and financial intermediation. *Journal of Finance*, 49(1):57–79.
- Chodorow-Reich, G. and Falato, A. (2022). The loan covenant channel: How bank health transmits to the real economy. *Journal of Finance*, 77(1):85–128.
- Court of the Southern District of New York (2014). *Marblegate Asset Management v. Education Management Finance Corp.* Opinion and Order No 14. Civ.8584 (KPF), from Dec 30.
- Datta, S., Iskandar-Datta, M., and Patel, A. (1999). Bank monitoring and the pricing of corporate public debt. *Journal of Financial Economics*, 51(3):435–449.

- De Fiore, F. and Uhlig, H. (2011). Bank finance versus bond finance. *Journal of Money, Credit and Banking*, 43(7):1399–1421.
- Demiroglu, C. and James, C. (2015). Bank loans and troubled debt restructurings. *Journal of Financial Economics*, 118(1):192–210.
- Denis, D. J. and Wang, J. (2014). Debt covenant renegotiations and creditor control rights. *Journal of Financial Economics*, 113(3):348–367.
- Diamond, D. W. (1984). Financial intermediation and delegated monitoring. *Review of Economic Studies*, 51(3):393–414.
- Diamond, D. W. (1991). Monitoring and Reputation: The Choice between Bank Loans and Directly Placed Debt. *Journal of Political Economy*, 99(4):689–721.
- Diamond, D. W. and Dybvig, P. H. (1983). Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91:401–419.
- Duffie, D. and Singleton, K. J. (1999). Modeling term structures of defaultable bonds. *Review of Financial Studies*, 12(4):687–720.
- Fama, E. F. (1985). What’s different about banks? *Journal of Monetary Economics*, 15(1):29–39.
- Gertner, R. and Scharfstein, D. (1991). A theory of workouts and the effects of reorganization law. *Journal of Finance*, 46(4):1189–1222.
- Gilson, S. C., John, K., and Lang, L. H. (1990). Troubled debt restructurings. An empirical study of private reorganization of firms in default. *Journal of Financial Economics*, 27(2):315–353.

- Glover, B. (2016). The expected cost of default. *Journal of Financial Economics*, 119(2):284–299.
- Gorton, G. and Kahn, J. (2000). The design of bank loan contracts. *Review of Financial Studies*, 13(2):331–364.
- Hackbarth, D., Hennessy, C. A., and Leland, H. E. (2007). Can the trade-off theory explain debt structure? *Review of Financial Studies*, 20(5):1389–1428.
- Hart, O. and Moore, J. (1988). Incomplete contracts and renegotiation. *Econometrica*, 56(4):755–785.
- Hoshi, T., Kashyap, A., and Scharfstein, D. (1990). The role of banks in reducing the costs of financial distress in Japan. *Journal of Financial Economics*, 27(1):67–88.
- James, C. (1987). Some evidence on the uniqueness of bank loans. *Journal of Financial Economics*, 19(2):217–235.
- Kornejew, M. (2025). Market creditor protection, finance and investment. *Working Paper*.
- Leland, H. E. and Pyle, D. H. (1977). Informational asymmetries, financial structure, and financial intermediation. *Journal of Finance*, 32(2):371–387.
- Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *Journal of Finance*, 29(2):449–470.
- Morellec, E., Valta, P., and Zhdanov, A. (2015). Financing investment: The choice between bonds and bank loans. *Management Science*, 61(11):2580–2602.
- Ramakrishnan, R. T. and Thakor, A. V. (1984). Information reliability and a theory of financial intermediation. *Review of Economic Studies*, 51(3):415–432.

- Roberts, M. R. (2015). The role of dynamic renegotiation and asymmetric information in financial contracting. *Journal of Financial Economics*, 116(1):61–81.
- Roberts, M. R. and Sufi, A. (2009). Renegotiation of financial contracts: Evidence from private credit agreements. *Journal of Financial Economics*, 93(2):159–184.
- Schwert, M. (2020). Does borrowing from banks cost more than borrowing from the market? *Journal of Finance*, 75(2):905–947.

Appendices

A.1 Pure Bank-Debt Case

The main text derives the benchmark debt rate y^* and the renegotiable loan rate y_l^* . Here I focus on the borrower side and derive the condition under which shareholders prefer a renegotiable loan. This yields the threshold $\bar{C}$ reported in Equation 5.

Shareholder condition and the renegotiation threshold Figure A.1 illustrates the shareholder payoff with and without renegotiation. Without renegotiation, shareholders repay the benchmark debt in the solvent state and receive zero in distress. With renegotiation, shareholders pay the renegotiable loan rate but preserve the option to continue in distress and recover at date 2.

Let V_2 denote firm value at date 2 in the recovery state. The difference in equity value between the renegotiation and no-renegotiation cases is

$$\Delta E = (1 - \rho_0)(y^* - y_l^*) + \rho_0(1 - \rho_1^L)(V_2 - 1). \quad (\text{A.1})$$

The first term captures the extra debt cost paid in solvent states when the renegotiable loan carries a higher rate. The second term captures the equity value preserved when renegotiation avoids immediate liquidation and the firm recovers.

Shareholders prefer renegotiation when $\Delta E > 0$, or equivalently,

$$\rho_0(1 - \rho_1^L)(V_2 - 1) > (1 - \rho_0)(y_l^* - y^*). \quad (\text{A.2})$$

This inequality makes the tradeoff explicit: the borrower accepts a higher rate ex ante only if

the expected continuation value in distress is sufficiently large. Solving $\Delta E = 0$ for the highest acceptable renegotiable loan rate yields

$$\bar{y}_l^* = y^* + \frac{\rho_0(1 - \rho_1^L)(V_2 - 1)}{1 - \rho_0}. \quad (\text{A.3})$$

Using the loan-rate expression in Equation 4, the indifference condition implies that the maximum renegotiation cost compatible with renegotiation is

$$\bar{C} = (1 - \rho_1^L)(\alpha + V_2 - 1). \quad (\text{A.4})$$

Hence, renegotiation is feasible if and only if $C < \bar{C}$. This is the threshold reported in the main text.

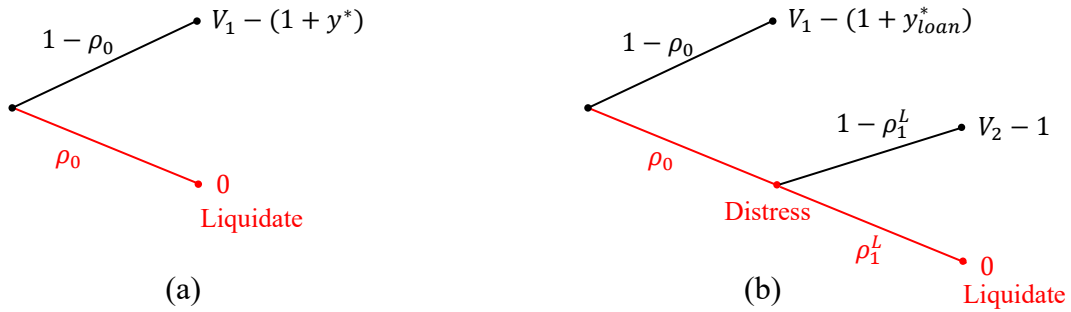


Figure A.1: The shareholder's payoff without renegotiation (Panel (a)) and with an extension option (Panel (b)). The borrower's asset value at time t is denoted as V_t . Without renegotiation, the shareholder repays the benchmark debt in the solvent state and receives zero in default. With renegotiation, the shareholder pays the renegotiable loan rate and benefits from continuation in distress if the firm recovers at time 2.

This interpretation can also be shown directly from firm asset values. Let ΔV denote the

difference in total firm value with and without renegotiation. Then

$$\begin{aligned}
\Delta V &= (1 - \rho_0)(V_1 - V_1) + \rho_0(1 - \rho_1^L)V_2 + \rho_0\rho_1^L(1 - \alpha) - \rho_0(1 - \alpha) \\
&= \rho_0 \left[\alpha(1 - \rho_1^L) + (1 - \rho_1^L)(V_2 - 1) \right] \\
&= \rho_0 \bar{C}.
\end{aligned} \tag{A.5}$$

Thus, $\bar{C}$ is exactly the per-unit surplus generated by renegotiation in the distress state.

A.2 Mixed-Borrowing Case

I now consider a firm that borrows from both a relationship bank and the bond market. The bank loan is senior and has face value $1 - F_b$, while the bond is junior and has face value F_b , with total face value normalized to 1. I assume $F_b < \alpha$, so in liquidation the junior bond is fully wiped out while the senior loan still bears default risk. This subsection proceeds in three steps. First, it derives the benchmark rates without renegotiation and shows how the seniority adjustment removes the mechanical spread difference. Second, it derives the loan premium when bondholders coordinate. Third, it introduces hold-out risk and derives the feasibility condition and equilibrium premium.

Benchmark rates and the seniority adjustment Figure A.2 illustrates the repayment structure of the senior loan and junior bond. Without renegotiation, the break-even conditions are

$$1 - F_b = (1 - F_b)(1 - \rho_0)(1 + y) + \rho_0(1 - \alpha), \tag{A.6}$$

$$F_b = F_b(1 - \rho_0)(1 + y). \tag{A.7}$$

Solving for the benchmark rates gives

$$y_l^* = \frac{(\alpha - F_b)\rho_0}{(1 - F_b)(1 - \rho_0)}, \quad y_b^* = \frac{\rho_0}{1 - \rho_0}. \quad (\text{A.8})$$

The bond rate exceeds the loan rate because the bond is junior. More precisely,

$$\frac{y_b^*}{y_l^*} = \frac{1 - F_b}{\alpha - F_b} = \frac{LGD_{bond}}{LGD_{loan}}, \quad (\text{A.9})$$

where $LGD_{bond} = 1$ and $LGD_{loan} = \frac{\alpha - F_b}{1 - F_b}$. Thus, the raw spread difference between loans and bonds mechanically reflects seniority. In the main text, the bond spread is therefore adjusted by the loss-given-default ratio before defining the loan premium.

Renegotiation when bondholders coordinate Suppose first that all bondholders coordinate with the bank and the firm, so there is no hold-out problem. In Figure A.2, this corresponds to the case $h = 0$ in the renegotiation panels. The break-even conditions become

$$1 - F_b = (1 - F_b)(1 - \rho_0)(1 + y) + \rho_0 \left[(1 - \rho_1^L)(1 - F_b) + \rho_1^L(1 - \alpha) - C \right], \quad (\text{A.10})$$

$$F_b = F_b(1 - \rho_0)(1 + y) + F_b\rho_0(1 - \rho_1^L). \quad (\text{A.11})$$

The corresponding rates are

$$y_l^0 = \frac{(\alpha - F_b)\rho_0\rho_1^L + \rho_0 C}{(1 - F_b)(1 - \rho_0)}, \quad y_b^0 = \frac{\rho_0\rho_1^L}{1 - \rho_0}. \quad (\text{A.12})$$

Using the seniority adjustment from Equation A.9, the adjusted bond rate is

$$y_{b,adj}^0 = y_b^0 \frac{LGD_{loan}}{LGD_{bond}} = \frac{(\alpha - F_b)\rho_0\rho_1^L}{(1 - F_b)(1 - \rho_0)}. \quad (\text{A.13})$$

Hence, the loan premium in the no-holdout case is

$$\pi_L^0 = y_l^0 - y_{b,adj}^0 = \frac{\rho_0 C}{(1 - F_b)(1 - \rho_0)}. \quad (\text{A.14})$$

The premium is increasing in C and in F_b . Renegotiation creates value for the whole debt structure, but the bank is compensated only through the senior loan tranche.

Shareholder condition without hold-out risk To derive the shareholder condition, compare total debt cost with and without renegotiation. Without renegotiation, total debt cost is

$$y_{total} = (1 - F_b)y_l^* + F_b y_b^* = \frac{\alpha \rho_0}{1 - \rho_0} = y^*. \quad (\text{A.15})$$

With renegotiation and no hold-out problem, total debt cost becomes

$$y_{total,reneg} = (1 - F_b)y_l^0 + F_b y_b^0 = \frac{\alpha \rho_0 \rho_1^L + \rho_0 C}{1 - \rho_0} = y_l^0. \quad (\text{A.16})$$

Therefore, the shareholder's incremental payoff is

$$\Delta E = (1 - \rho_0)(y_{total} - y_{total,reneg}) + \rho_0(1 - \rho_1^L)(V_2 - 1). \quad (\text{A.17})$$

The condition $\Delta E > 0$ again implies the same threshold as in the pure bank case:

$$C < \bar{C}. \quad (\text{A.18})$$

Thus, in the absence of hold-outs, mixed borrowing changes the allocation of payoffs across claims, but it does not change the renegotiation threshold.

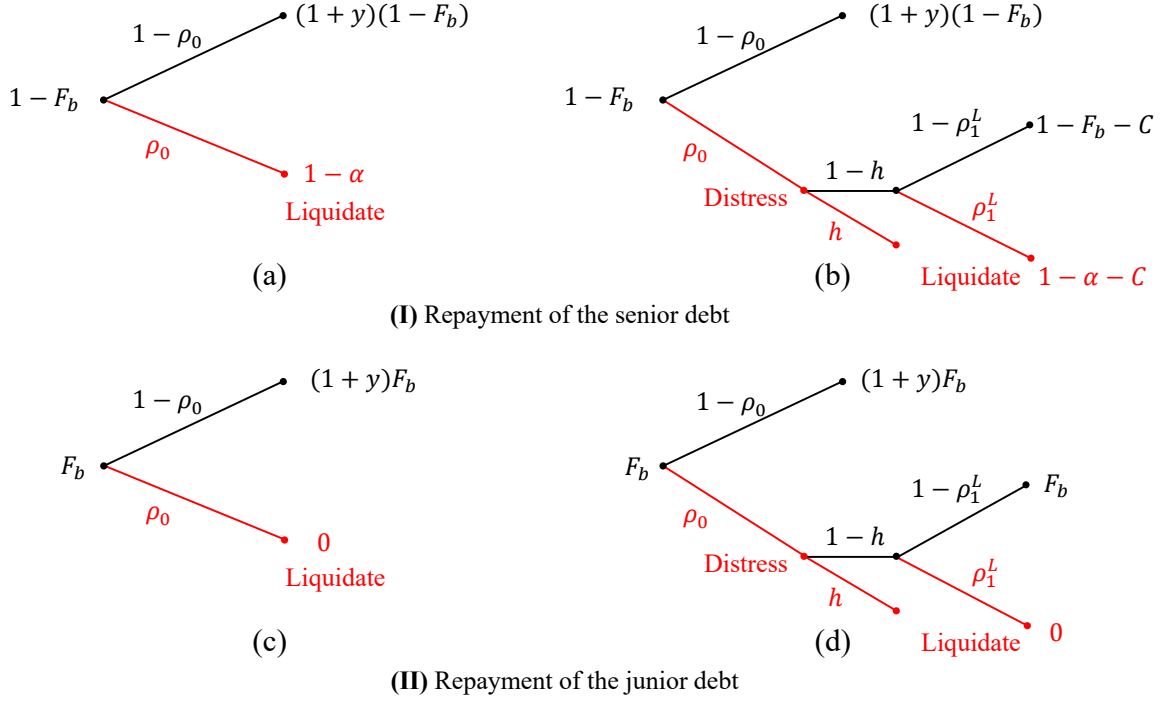


Figure A.2: Repayment structures for mixed borrowing. Panel (I) illustrates the repayment of the senior debt (loan), and Panel (II) shows the repayment of the junior debt (bond). The face values of loan and bond sum to 1, and the bond face value is denoted as F_b .

Pricing with hold-out risk I now introduce hold-out risk in the same reduced-form way as in the main text. Suppose that when the bank attempts a restructuring in distress, renegotiation fails with probability h and succeeds with probability $1 - h$, where a higher h reflects more severe hold-out frictions. If renegotiation succeeds, the firm continues to date 2 and defaults there with probability ρ_1^L ; if renegotiation fails, the firm is liquidated immediately. Under this formulation, the senior loan and junior bond satisfy the following break-even conditions:

$$1 - F_b = (1 - F_b)(1 - \rho_0)(1 + y) + \rho_0 \left[(1 - h) \left((1 - \rho_1^L)(1 - F_b) + \rho_1^L(1 - \alpha) \right) + h(1 - \alpha) - C \right], \quad (\text{A.19})$$

$$F_b = F_b(1 - \rho_0)(1 + y) + \rho_0(1 - h)(1 - \rho_1^L)F_b. \quad (\text{A.20})$$

Solving these equations gives

$$y_l^h = \frac{(\alpha - F_b)\rho_0[h + (1 - h)\rho_1^L] + \rho_0 C}{(1 - F_b)(1 - \rho_0)}, \quad y_b^h = \frac{\rho_0[h + (1 - h)\rho_1^L]}{1 - \rho_0}. \quad (\text{A.21})$$

Conditional loan premium The seniority-adjusted bond rate under hold-out risk is

$$y_{b,adj}^h = y_b^h \frac{LGD_{loan}}{LGD_{bond}} = \frac{(\alpha - F_b)\rho_0[h + (1 - h)\rho_1^L]}{(1 - F_b)(1 - \rho_0)}. \quad (\text{A.22})$$

Therefore, conditional on renegotiation taking place, the loan premium is

$$\pi_L^h = y_l^h - y_{b,adj}^h = \frac{\rho_0 C}{(1 - F_b)(1 - \rho_0)}. \quad (\text{A.23})$$

This derivation shows why the main-text premium formula does not depend on h conditional on renegotiation. Hold-out risk affects loan and bond payoffs symmetrically through the timing of default, and this common component disappears once the bond spread is adjusted for seniority. Thus, hold-out risk does not affect the premium directly conditional on renegotiation; instead, it affects whether renegotiation remains feasible.

Shareholder condition with hold-out risk Without renegotiation, total debt cost remains

$$y_{total} = \frac{\alpha \rho_0}{1 - \rho_0}. \quad (\text{A.24})$$

With renegotiation and hold-out risk, total debt cost is

$$\begin{aligned} y_{total,reneg}(h) &= (1 - F_b)y_l^h + F_b y_b^h \\ &= \frac{\rho_0 [\alpha(h + (1 - h)\rho_1^L) + C]}{1 - \rho_0}. \end{aligned} \quad (\text{A.25})$$

Hence the shareholder's incremental payoff is

$$\Delta E(h) = (1 - \rho_0) [y_{total} - y_{total,reneg}(h)] + \rho_0(1 - h)(1 - \rho_1^L)(V_2 - 1). \quad (\text{A.26})$$

Relative to the no-holdout case, the continuation value is now multiplied by $1 - h$, because successful restructuring occurs less often. Substituting the two debt-cost expressions and simplifying gives

$$\Delta E(h) = \rho_0 \left[(1 - h)(1 - \rho_1^L)(\alpha + V_2 - 1) - C \right]. \quad (\text{A.27})$$

Therefore, renegotiation is attractive if and only if

$$C < \bar{C}_h, \quad \bar{C}_h = (1 - h)(1 - \rho_1^L)(\alpha + V_2 - 1) = (1 - h)\bar{C}. \quad (\text{A.28})$$

The threshold falls with hold-out risk because renegotiation succeeds less often and therefore preserves continuation value in fewer states.

Equilibrium premium Combining the conditional pricing result with the feasibility condition, the equilibrium premium can be written as

$$\pi_L = \begin{cases} \frac{\rho_0 C}{(1 - F_b)(1 - \rho_0)}, & \text{if } C < \bar{C}_h, \\ 0, & \text{if } C \geq \bar{C}_h. \end{cases} \quad (\text{A.29})$$

This piecewise expression makes clear that hold-out risk operates mainly through an extensive-margin channel. It shrinks the set of firms for which renegotiation is feasible and can therefore eliminate the premium altogether, even though it does not affect the conditional premium in Equation A.23.

If hold-out risk is increasing in bond reliance, $h = h(F_b)$ with $h'(F_b) > 0$, then bond-intensive firms are more exposed to this extensive-margin effect. A sufficiently large increase in h can push those firms from the renegotiation region into the no-renegotiation region, causing the premium to disappear.

A.3 Alternative Repayment Assumption at Date 2

The main analysis assumes that if the firm recovers after renegotiation, the bank receives only the principal at date 2. This simplification keeps the algebra transparent. I now show that the key implications remain unchanged if the lender instead receives both principal and interest in the recovery state.

The benchmark rate without renegotiation remains

$$y^* = \frac{\alpha \rho_0}{1 - \rho_0}. \quad (\text{A.30})$$

If the lender receives full face value plus interest after successful restructuring, the break-even condition becomes

$$1 = (1 - \rho_0)(1 + y) + \rho_0 [\rho_1^L(1 - \alpha) + (1 - \rho_1^L)(1 + y) - C], \quad (\text{A.31})$$

which implies

$$y_l^* = \frac{\alpha \rho_0 \rho_1^L + \rho_0 C}{1 - \rho_0 \rho_1^L}. \quad (\text{A.32})$$

The shareholder's payoff difference becomes

$$\Delta E = (1 - \rho_0)(y^* - y_l^*) + \rho_0(1 - \rho_1^L) [V_2 - (1 + y_l^*)]. \quad (\text{A.33})$$

Solving $\Delta E = 0$ again yields the same threshold as in Equation A.4. Therefore, the main qualitative results are robust to this alternative assumption.

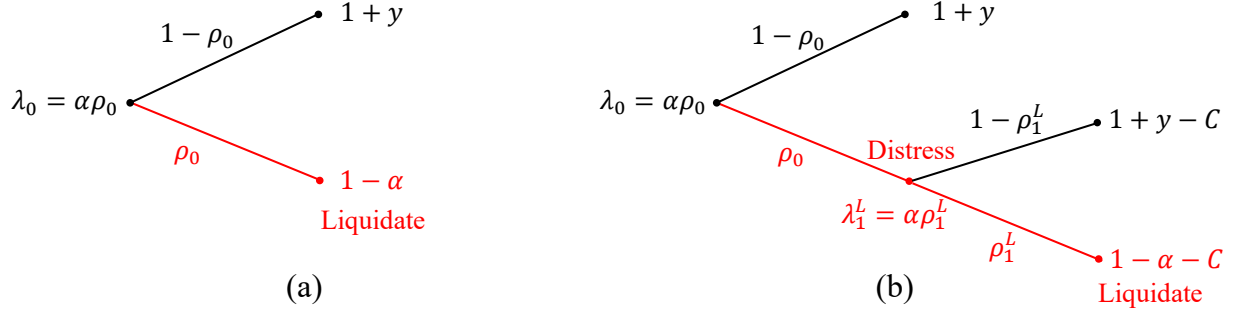


Figure A.3: The one-period debt (Panel (a)) and the debt with an extension in distress (Panel (b)). Here, the lender receives both principal and interest at date 2 if the firm recovers after renegotiation.

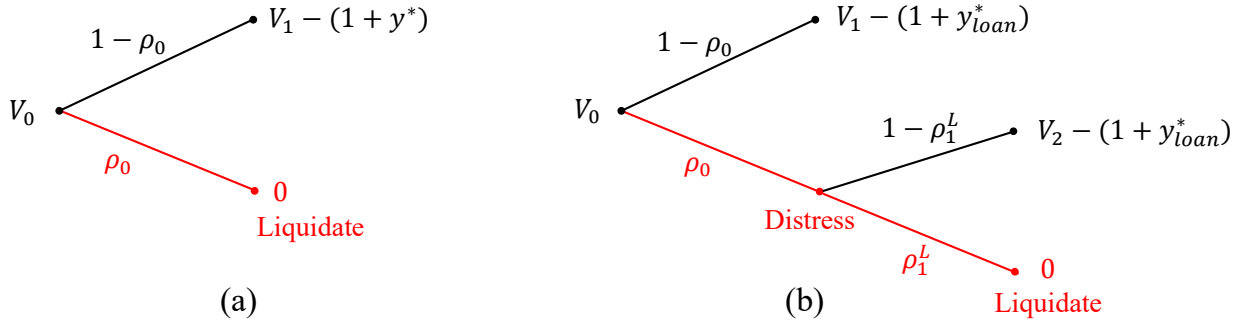


Figure A.4: The shareholder's payoff without renegotiation (Panel (a)) and with an extension option (Panel (b)) when the lender receives both principal and interest after recovery.

A.4 Prepayment Option

To explore the dynamics of the prepayment option, I consider a two-period loan contract. This setup captures how credit risk evolves over time and the strategic interactions between borrower and lender. The payoff of the debt contract is depicted in Figure A.5.

Assume the state variable λ represents the short-term credit risk of the borrower and that there is a constant loss given default α , the face value of debt is normalized to 1. The repayment

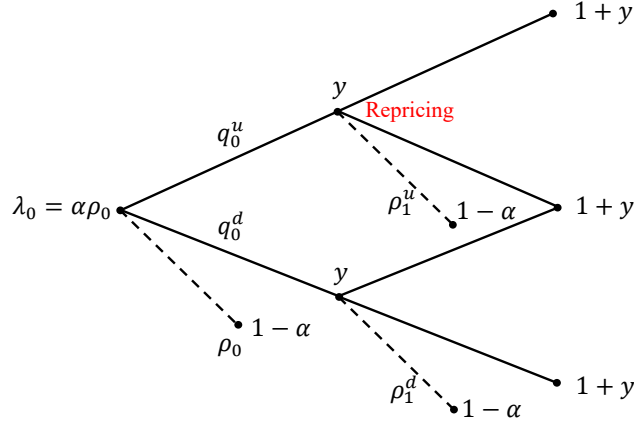


Figure A.5: The two-period debt with a prepayment option. The variable λ denotes the short-term credit risk of a firm. In the up-state with a probability of q_t^u , the firm's credit quality improves, and in the down-state with a probability of q_t^d , the firm's credit quality deteriorates. The branches plotted in dashed lines denote the default states.

schedule is as follows: at time $t = 1$, the firm repays a pre-specified interest rate y , and at time $t = 2$, it repays both the principal and the interest, $1 + y$. If the firm defaults with a probability ρ_t , it is liquidated, and the bank recovers a fraction $(1 - \alpha)$ of the face value. If the firm remains solvent, the credit quality may either improve (up-state) or deteriorate (down-state), with associated risk-neutral probabilities q_t^u and q_t^d , respectively. At each node, the probabilities sum to one,

$$q_t^u + q_t^d + \rho_t = 1. \quad (\text{A.34})$$

The fair coupon rate for the debt, when traded at par, can be calculated using:

$$1 = y(1 - \rho_0) + (1 - \alpha)\rho_0 + (1 + y)[q_0^u(1 - \rho_1^u) + q_0^d(1 - \rho_1^d)] + (1 - \alpha)[q_0^u\rho_1^u + q_0^d\rho_1^d]. \quad (\text{A.35})$$

Solving for y , and denoting the solution as y_2^* ,

$$y_2^* = \frac{\alpha\rho_0 + \alpha(q_0^u\rho_1^u + q_0^d\rho_1^d)}{(1 - \rho_0) + q_0^u(1 - \rho_1^u) + q_0^d(1 - \rho_1^d)}. \quad (\text{A.36})$$

Next, I consider a scenario in which the bank anticipates that, if the firm's credit quality improves at time 1, the firm will request a repricing of the second-period interest rate. Since the borrower can prepay the loan at a low cost, it has the bargaining power to terminate the existing contract and initiate a new one if the bank disagrees with the requested rate adjustment. This repricing only occurs if the firm's credit quality improves in the up-state at time 1, allowing the borrower to demand a lower interest rate for the second period. Conversely, if the firm's credit deteriorates, it retains the original contract, which is preferable. This self-selection feature aligns with the findings of [Roberts and Sufi \(2009\)](#) and [Roberts \(2015\)](#). Anticipating the repricing activity, banks will require compensation for the prepayment and repricing risk. To determine the value of this compensation, I first derive the adjusted yield requested by the borrower in the up-state, denoted as y_u . I define y_u as the fair yield of a one-period debt conditional on arriving in the up-state at time $t = 1$, which is equivalent to the firm terminating the original debt and initiating a new one. The adjusted yield must satisfy:

$$1 = (1 + y)(1 - \rho_1^u) + (1 - \alpha)\rho_1^u. \quad (\text{A.37})$$

The solution is,

$$y_u = \frac{\alpha\rho_1^u}{1 - \rho_1^u}. \quad (\text{A.38})$$

I show that y_u is always smaller than the original fair yield y_2^* (see the following mathematical proof). The adjusted yield y_u is lower because it only accounts for the reduced credit risk after the firm's credit health has improved, whereas y_2^* also accounts for the worse outcomes in the down-state. Given the requested new yield, the value of the prepayment option, P , is

$$P = q_0^u(y_2^* - y_u)(1 - \rho_1^u). \quad (\text{A.39})$$

It represents the difference between the original fair yield and the adjusted yield that the bank can receive at time 2, conditional on the firm reaching the up-state at time 1 and remaining solvent at time 2. The option value would increase with q_0^u , and decrease with ρ_1^u (or λ_1^u) since the multiplier $y_2^* - y_u$ is positive. This intuition makes sense: when it is more likely to reach the up-state, the firm is more likely to exercise the option, making the prepayment option more valuable. Similarly, when the firm's credit quality gets a larger improvement (ρ_1^u decreases), the prepayment option value increases as the adjusted yield becomes even more attractive.

Mathematically, I find that while the option value indeed decreases with λ_1^u ($\frac{\partial P}{\partial \lambda_1^u} < 0$), the relationship with q_0^u is more complex. To examine the dynamics of the option value with q_0^u , I solve the first-order condition $\frac{\partial P}{\partial q_0^u} = 0$, and identify the value of q_0^{u*} that maximizes P ,

$$q_0^{u*} = \frac{-(1 - \rho)(2 - \rho_1^d) + \sqrt{(1 - \rho)(2 - \rho)(2 - \rho_1^d)(1 - \rho_1^u)}}{\rho_1^d - \rho_1^u}. \quad (\text{A.40})$$

By making very mild assumptions on the parameters, I approximate that q_0^{u*} is in the range of 0.7 to 0.83 (see the following mathematical proof). So, the option value first increases with q_0^u , but peaks when q_0^u reaches a threshold of around 70% and then starts to decrease. Intuitively, the option price increases in the probability q_0^u as it becomes more likely for the firm to exercise the option. Simultaneously, the original yield y_2^* decreases as q_0^u rises. Once q_0^u reaches the threshold, the yield difference between the original and renewed contracts becomes very small, which makes the prepayment option less attractive, explaining why the option value starts to decrease thereafter. However, this threshold of 0.7 is much larger than the realistic value of q_0^u that is typically assumed to be around 0.5. Thus, I conclude that in most cases, P increases with q_0^u .

I also find that the prepayment option price increases with λ_0 . Since q_0^u , y_u , and λ_1^u are

uncorrelated with λ_0 , the option price depends on λ_0 only through y_2^* . Thus, $\frac{\partial P}{\partial \lambda_0}$ has the same sign as $\frac{\partial y_2^*}{\partial \lambda_0}$, which is positive.

To derive the price of the option as a function of the debt's coupon, I add the prepayment option value to the left-hand side of Equation A.35:

$$1 + P = y(1 - \rho_0) + (1 - \alpha)\rho_0 + (1 + y)[q_0^u(1 - \rho_1^u) + q_0^d(1 - \rho_1^d)] + (1 - \alpha)[q_0^u\rho_1^u + q_0^d\rho_1^d]. \quad (\text{A.41})$$

Solving for the yield that a bank should charge ex ante to account for the prepayment risk, and denoting the solution as y_p^* ,

$$y_p^* = y_2^* + \frac{P}{\underbrace{(1 - \rho_0) + q_0^u(1 - \rho_1^u) + q_0^d(1 - \rho_1^d)}_{\text{Prepayment premium}}}. \quad (\text{A.42})$$

Calibration It turns out that the prepayment compensation term is very small compared to y_2^* . Based on a calibration using parameters of $\rho_0 = 4\%$, $\rho_1^u = 2.4\%$, $\rho_1^d = 6.7\%$, $\alpha = 50\%$, $q_0^u = 50\%$, $q_0^d = 46\%$, I find a fair yield y_2^* of 245 bps, with a prepayment compensation of only 25 bps. The small size of this adjustment is also evident from the prepayment option value in Equation A.39. The value is primarily determined by the yield difference between y_2^* and y_u , which is minimal. Consequently, it is not surprising that the compensation for the prepayment option is a very small fraction of the loan's fair yield.

The analysis shows that while prepayment flexibility is a distinct advantage of bank loans, its contribution to the overall loan premium is small. Furthermore, it remains an empirical question whether prepayment risk is incorporated into the interest rate, thereby contributing to the loan premium. If prepayment risk is priced into the loan premium, the model suggests that the premium will increase when there is a higher probability of the firm's credit quality

improving; and when the expected improvement is larger.

Mathematical proof in the prepayment option The value of the prepayment option P , the initial interest rate without prepayment option y_2^* , and the requested new interest rate y_r^* are given by,

$$P = q_0^u(y_2^* - y_u)(1 - \rho_1^u),$$

$$y_2^* = \frac{\alpha\rho_0 + \alpha(q_0^u\rho_1^u + q_0^d\rho_1^d)}{(1 - \rho_0) + q_0^u(1 - \rho_1^u) + q_0^d(1 - \rho_1^d)}, \quad \text{and} \quad y_u = \frac{\alpha\rho_1^u}{1 - \rho_1^u}.$$

Denote $\hat{\rho} = q_0^u\rho_1^u + q_0^d\rho_1^d$, y_2^* can be rewritten as,

$$y_2^* = \frac{\alpha\rho_0 + \alpha\hat{\rho}}{2(1 - \rho_0) - \hat{\rho}}.$$

Prove that $y_2^* > y_u$:

$$\begin{aligned} y_2^* - y_u &= \frac{(\alpha\rho_0 + \alpha\hat{\rho})(1 - \rho_1^u) - \alpha\rho_1^u[2(1 - \rho_0) - \hat{\rho}]}{[2(1 - \rho_0) - \hat{\rho}](1 - \rho_1^u)} \\ &= \alpha \frac{\rho_0 + \hat{\rho} + \rho_0\rho_1^u - 2\rho_1^u}{[2(1 - \rho_0) - \hat{\rho}](1 - \rho_1^u)} \\ &= \alpha \frac{(\rho_0 - \rho_1^u) + (\hat{\rho} + \rho_0\rho_1^u - \rho_1^u)}{[2(1 - \rho_0) - \hat{\rho}](1 - \rho_1^u)} \end{aligned} \tag{A.43}$$

I show that $\hat{\rho} + \rho_0\rho_1^u - \rho_1^u = q_0^u\rho_1^u + q_0^d\rho_1^d + \rho_0\rho_1^u - \rho_1^u = q_0^d(\rho_1^d - \rho_1^u) > 0$. Since both the value of $\rho_0 - \rho_1^u$ and $\hat{\rho} + \rho_0\rho_1^u - \rho_1^u$ are positive, Equation A.43 is positive since the denominator is also positive.

Prove that P decreases in ρ_1^u : Since λ_1^u is independent of q_0^u , I only need to examine the partial derivative of $\tilde{P} \equiv (y_2^* - y_u)(1 - \rho_1^u)$ with respect to ρ_1^u .

$$\begin{aligned}\tilde{P} &= (y_2^* - y_u)(1 - \rho_1^u) \\ &= \alpha \frac{\rho_0 + \hat{\rho} + \rho_0 \rho_1^u - 2\rho_1^u}{[2(1 - \rho_0) - \hat{\rho}]} \\ &= \alpha \left(\frac{(2 - \rho_0)(1 - \rho_1^u)}{[2(1 - \rho_0) - \hat{\rho}]} - 1 \right).\end{aligned}\tag{A.44}$$

The partial derivative is,

$$\begin{aligned}\frac{\partial \tilde{P}}{\partial \rho_1^u} &= \alpha \frac{-(2 - \rho_0)[2(1 - \rho_0) - \hat{\rho}] + (2 - \rho_0)(1 - \rho_1^u)q_0^u}{[2(1 - \rho_0) - \hat{\rho}]^2} \\ &= \alpha \frac{(2 - \rho_0)[-2(1 - \rho_0) + q_0^u \rho_1^u + q_0^d \rho_1^d + (1 - \rho_1^u)q_0^u]}{[2(1 - \rho_0) - \hat{\rho}]^2} \\ &= \alpha \frac{(2 - \rho_0)[-2(1 - \rho_0) + q_0^u + q_0^d \rho_1^d]}{[2(1 - \rho_0) - \hat{\rho}]^2} \\ &= \alpha \frac{(2 - \rho_0)[-2(1 - \rho_0) + 1 - \rho_0 - q_0^d + q_0^d \rho_1^d]}{[2(1 - \rho_0) - \hat{\rho}]^2} \\ &= \alpha \frac{(2 - \rho_0)[-(1 - \rho_0) - q_0^d(1 - \rho_1^d)]}{[2(1 - \rho_0) - \hat{\rho}]^2} < 0.\end{aligned}\tag{A.45}$$

Prove that y_2^* is decreasing in q_0^u : Since $\frac{\partial \hat{\rho}}{\partial q_0^u} = \rho_1^u - \rho_1^d < 0$, and

$$\begin{aligned}\frac{\partial y_2^*}{\partial q_0^u} &= \alpha \frac{(\rho_1^u - \rho_1^d)[2(1 - \rho_0) - \hat{\rho}] + (\rho_0 + \hat{\rho})(\rho_1^u - \rho_1^d)}{[2(1 - \rho_0) - \hat{\rho}]^2} \\ &= \alpha \frac{(\rho_1^u - \rho_1^d)[2(1 - \rho_0) - \hat{\rho} + \rho_0 + \hat{\rho}]}{[2(1 - \rho_0) - \hat{\rho}]^2} \\ &= \alpha \frac{(\rho_1^u - \rho_1^d)(2 - \rho_0)}{[2(1 - \rho_0) - \hat{\rho}]^2} < 0.\end{aligned}\tag{A.46}$$

To examine whether the option value P is increasing or decreasing in q_0^u (i.e. the sign of

$\frac{\partial P}{\partial q_0^u}$), I define $\hat{P} \equiv q_0^u(y_2^* - y_u)$ since q_0^u is independent of $(1 - \lambda_1^u)$ and $(1 + r)^2$, so,

$$\begin{aligned} \frac{\partial \hat{P}}{\partial q_0^u} &= (y_2^* - y_u) + q_0^u \frac{\partial y_2^*}{\partial q_0^u} \\ &= \alpha \frac{(\rho_0 - \rho_1^u) + q_0^d(\rho_1^d - \rho_1^u)}{[2(1 - \rho_0) - \hat{\rho}](1 - \rho_1^u)} + q_0^u \alpha \frac{(\rho_1^u - \rho_1^d)(2 - \rho_0)}{[2(1 - \rho_0) - \hat{\rho}]^2}. \end{aligned} \quad (\text{A.47})$$

Denote $\epsilon = \rho_1^d - \rho_1^u$,

$$\frac{\partial \hat{P}}{\partial q_0^u} = \alpha \frac{(\rho_0 - \rho_1^u) + q_0^d \epsilon}{[2(1 - \rho_0) - \hat{\rho}](1 - \rho_1^u)} + q_0^u \alpha \frac{-\epsilon(2 - \rho_0)}{[2(1 - \rho_0) - \hat{\rho}]^2}. \quad (\text{A.48})$$

The first-order condition is,

$$\frac{\partial \hat{P}}{\partial q_0^u} = \alpha \frac{(\rho_0 - \rho_1^u) + q_0^d \epsilon}{[2(1 - \rho_0) - \hat{\rho}](1 - \rho_1^u)} - q_0^u \alpha \frac{\epsilon(2 - \rho_0)}{[2(1 - \rho_0) - \hat{\rho}]^2} = 0, \quad (\text{A.49})$$

$$\frac{(\rho_0 - \rho_1^u) + (1 - \rho_0 - q_0^u) \epsilon}{(1 - \rho_1^u)} = \frac{q_0^u \epsilon (2 - \rho_0)}{[2(1 - \rho_0) + q_0^u \epsilon - (1 - \rho_0) \rho_1^d]}. \quad (\text{A.50})$$

Denote $\tilde{q} = q_0^u \epsilon$,

$$\frac{(\rho_0 - \rho_1^u) + (1 - \rho_0) \epsilon - \tilde{q}}{(1 - \rho_1^u)} = \frac{\tilde{q}(2 - \rho_0)}{[2(1 - \rho_0) + \tilde{q} - (1 - \rho_0) \rho_1^d]}. \quad (\text{A.51})$$

It is a quadratic function of $\tilde{q}$,

$$[(\rho_0 - \rho_1^u) + (1 - \rho_0) \epsilon - \tilde{q}][2(1 - \rho_0) - (1 - \rho_0) \rho_1^d + \tilde{q}] - \tilde{q}(1 - \rho_1^u)(2 - \rho_0) = 0. \quad (\text{A.52})$$

I rewrite it as,

$$\tilde{q}^2 + 2(1 - \rho)(2 - \rho_1^d) \tilde{q} - (1 - \rho)(2 - \rho_1^d)[(\rho_0 - \rho_1^u) + (1 - \rho) \epsilon] = 0. \quad (\text{A.53})$$

Roots for this quadratic function always exist because the discriminant is,

$$[2(1 - \rho)(2 - \rho_1^d)]^2 + r(1 - \rho)(2 - \rho_1^d)[(\rho_0 - \rho_1^u) + (1 - \rho)\epsilon] > 0.$$

The positive root is:

$$\begin{aligned} \tilde{q} &= -(1 - \rho)(2 - \rho_1^d) + \sqrt{[(1 - \rho)(2 - \rho_1^d)]^2 + (1 - \rho)(2 - \rho_1^d)[(\rho_0 - \rho_1^u) + (1 - \rho)\epsilon]} \\ &= -(1 - \rho)(2 - \rho_1^d) + \sqrt{(1 - \rho)(2 - \rho)(2 - \rho_1^d)(1 - \rho_1^u)}. \end{aligned} \quad (\text{A.54})$$

The corresponding root for q_0^u is:

$$q_0^{u*} = \frac{-(1 - \rho)(2 - \rho_1^d) + \sqrt{(1 - \rho)(2 - \rho)(2 - \rho_1^d)(1 - \rho_1^u)}}{\rho_1^d - \rho_1^u}. \quad (\text{A.55})$$

I further show that q_0^{u*} approaches 0.75 as ρ approaches zero. Proof is as follows. To approximate the quotient of two small values as in Equation A.55, I take the expression of the default probability at time 1 as, $\rho_1^u = \rho_0 - \beta$, $\rho_1^d = \rho_0 + \kappa\beta$. This does not imply that ρ_1^u and ρ_1^d are perfectly linear in ρ_0 ; rather, they are simply offset with respect to ρ_0 . Since ρ_0 is very small, $\beta \approx 0$, and the movement of the credit spread may be asymmetric if $\kappa \neq 1$ (with κ being positive). Assume the upward movement is not significantly different from the downward movement, κ should not deviate much from 1. Rewrite Equation A.55 as:

$$q_0^{u*} = \frac{(1 - \rho)(2 - \rho_1^d) \left[-1 + \sqrt{\frac{(2 - \rho)(1 - \rho_1^u)}{(1 - \rho)(2 - \rho_1^d)}} \right]}{(1 + \kappa)\beta}. \quad (\text{A.56})$$

A linear approximation of $-1 + \sqrt{\frac{(2-\rho)(1-\rho_1^u)}{(1-\rho)(2-\rho_1^d)}}$ around $\beta = 0$ is:

$$\begin{aligned} -1 + \sqrt{\frac{(2-\rho)(1-\rho_1^u)}{(1-\rho)(2-\rho_1^d)}} &\approx -1 + \sqrt{\frac{(2-\rho)(1-\rho)}{(1-\rho)(2-\rho)}} + \frac{\beta}{2} \sqrt{\frac{(2-\rho)(1-\rho)}{(1-\rho)(2-\rho)}} \frac{(2-\rho) + (1-\rho)\kappa}{(2-\rho)^2} \\ &= \frac{\beta}{2} \frac{(2-\rho) + (1-\rho)\kappa}{(2-\rho)(1-\rho)}. \end{aligned} \tag{A.57}$$

Thus,

$$q_0^{u*} \approx \frac{(1-\rho)(2-\rho)}{(1+\kappa)\beta} \frac{\beta}{2} \frac{(2-\rho) + (1-\rho)\kappa}{(2-\rho)(1-\rho)} = \frac{1 + (1-\rho)(1+\kappa)}{2(1+\kappa)}. \tag{A.58}$$

If κ is in the range from 0.5 to 1.5, then as λ_0 approaches zero, q_0^{u*} will be in the range between 0.7 and 0.83. If $\kappa = 1$, meaning the upward and downward movements are symmetric, then q_0^{u*} will approach 0.75.

A.5 Additional Figures and Tables

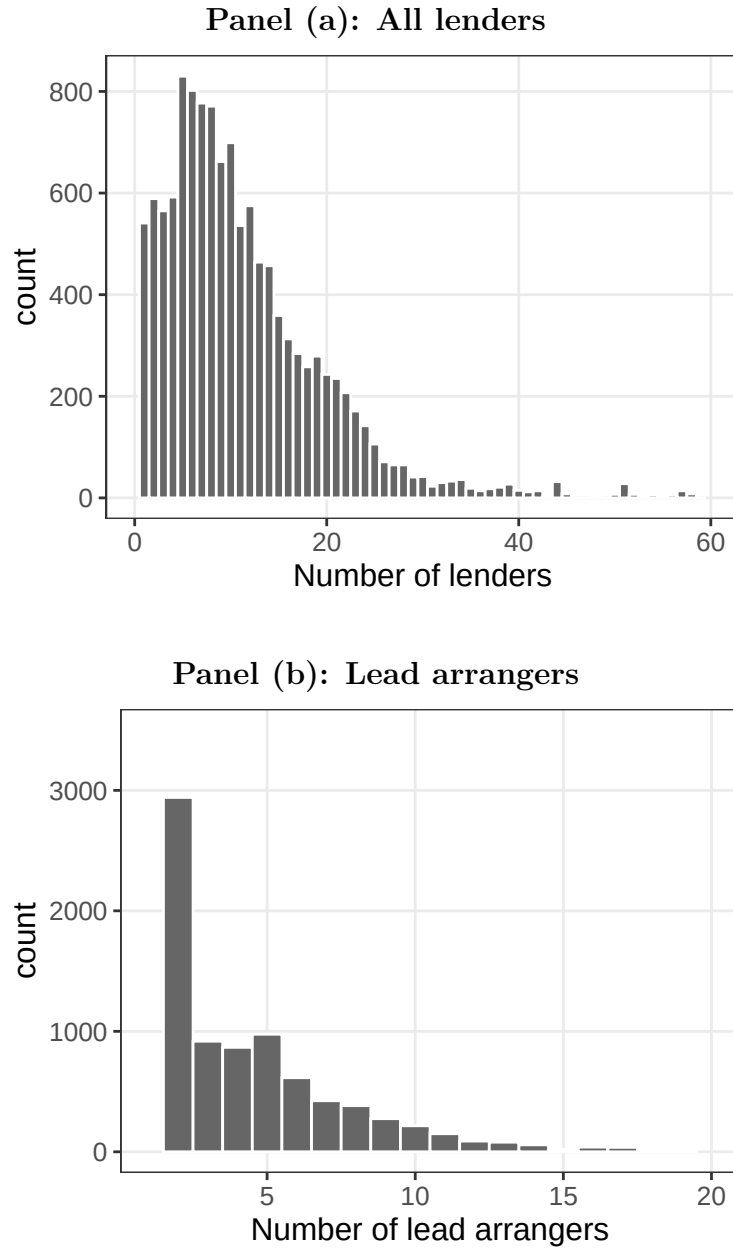


Figure A.6: Distribution of syndicate size in the final sample. Panel (a) shows the number of lenders in each loan syndicate, and Panel (b) shows the number of lead arrangers. The figure illustrates that most loans are syndicated rather than bilateral, which motivates using the number of lead arrangers as a proxy for renegotiation frictions in the heterogeneity analysis.

Table A.I: Mapping from S&P issuer ratings to numeric rating scores. This table reports the numeric scores assigned to historical S&P long-term issuer ratings in the empirical analysis. Lower scores correspond to better credit quality.

Value	S&P	Category
1	AAA	Investment Grade
2	AA+	
3	AA	
4	AA-	
5	A+	
6	A	
7	A-	
8	BBB+	
9	BBB	
10	BBB-	
11	BB+	Non-Investment Grade
12	BB	
13	BB-	
14	B+	
15	B	
16	B-	
17	CCC+	
18	CCC	
19	CCC-	
20	CC	
21	C	
25	D	Default

Table A.II: Notation and variable definitions. This table summarizes the main notation used in the model and the definitions of the key empirical variables.

Symbol/Term	Definition
<i>Modeling: renegotiation channel</i>	
α	Loss given default.
ρ_0	Distress probability at date 1.
ρ_1^L	Default probability at date 2 after renegotiation.
y^*	Benchmark rate in the pure bank-debt case.
$\underline{y}^*$	Lower bound on the renegotiated loan rate when $C = 0$.
y_l^*	Context-specific loan rate: renegotiated in pure bank debt, benchmark senior in mixed borrowing.
y_b^*	Benchmark junior bond rate in mixed borrowing.
$y_{b,adj}^*$	Seniority-adjusted benchmark bond rate in mixed borrowing.
y_l^0	Senior loan rate in mixed borrowing when $h = 0$.
y_b^0	Junior bond rate in mixed borrowing when $h = 0$.
$y_{b,adj}^0$	Seniority-adjusted bond rate in mixed borrowing when $h = 0$.
y_l^h	Senior loan rate in mixed borrowing with holdout risk.
y_b^h	Junior bond rate in mixed borrowing with holdout risk.
$y_{b,adj}^h$	Seniority-adjusted bond rate in mixed borrowing with holdout risk.
F_b	Bond face-value share in mixed borrowing.
V_t	Firm asset value at date t .
V_2	Firm value at date 2 in the recovery state.
C	Bank renegotiation cost.
$\bar{C}$	Max renegotiation cost without holdout risk.
$\bar{C}_h$	Max renegotiation cost with holdout risk.
h	Probability a restructuring attempt fails because of holdouts.
π_L^0	Loan premium in mixed borrowing when $h = 0$.
π_L	Equilibrium loan premium in mixed borrowing.
π_L^h	Conditional loan premium with holdout risk.
<i>Empirical measures</i>	
Book debt	Total long-term debt plus current liabilities.
Market asset	Book debt plus equity market capitalization.
Market leverage	Ratio of book debt to market asset.
Equity volatility	Annualized s.d. of daily stock returns over the prior year.
Asset volatility	Unlevered equity volatility.
Trailing stock ret	Firm's stock return over the prior year.
Subsequent stock ret	Firm's stock return over the following year.
Asset tangibility	Ratio of tangible asset to book assets.
Profitability	Ratio of operating income before depreciation to book assets.

Table A.III: Comparison of treated and untreated firms. This table compares summary statistics for firms with high and low reliance on bond financing in the pre-Marblegate period. The treated group consists of bond-intensive firms, defined as firms whose bond holdings scaled by book assets are above the sample median. The table reports the main loan, bond, and firm characteristics used in the empirical analysis.

	N		Mean		Median	
	Control	Treated	Control	Treated	Control	Treated
All-in-drawn spread (bps)	4279	4331	170	230	150	225
Tranche amount (\$M)	4279	4331	764	549	420	300
Loan Maturity	4279	4331	4.4	4.4	5	4.9
Yield to maturity (%)	4279	4331	6.5	8.7	5.7	7.1
Yield spread (bps)	4279	4331	323	536	179	354
Face value (\$M)	4279	4331	351	357	250	270
Bond Maturity	4279	4331	4.5	4.6	4.6	4.7
Market asset (\$B)	3484	3336	22	8.7	6.7	3.2
Market leverage (%)	3484	3336	37	48	35	46
Asset volatility	3439	3246	0.22	0.23	0.2	0.2
Distance to default	3438	3245	5.8	3.6	5.7	4.2
Trailing stock return (%)	3461	3265	13	23	7.3	4.1
Asset market to book (%)	3484	3336	111	122	94	101
Asset tangibility (%)	4099	4260	39	40	37	37
Profitability (%)	3961	4180	3	2.8	2.8	2.9
Bond to book asset (%)	4117	4331	15	50	16	39
S&P rating score	2983	3281	11	13	10	13
Reduced form (bps)	4279	4331	90	96	84	102
Structural form (bps)	1489	1703	33	71	67	125

Table A.IV: Heterogeneity in the Marblegate effect by secured status. This table reports triple-difference estimates of the response of the loan premium to the Marblegate ruling in the sample of loans originated through 2016. The dependent variable is the loan premium in basis points. The interaction *Marblegate* $\times$ *Bond intensive* captures the post-Marblegate effect for the omitted group of unsecured loans, while the triple interaction *Marblegate* $\times$ *Bond intensive* $\times$ *Secured indicator* captures the incremental effect for secured loans relative to unsecured loans. The secured indicator equals one if the loan is secured. Industry fixed effects are included in all specifications, and firm-level covariates are added in column (2). Standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>	
	Loan premium	
	(1)	(2)
Marblegate*Bond intensive	−42.30 (53.38)	−80.78* (45.54)
Marblegate*Bond intensive*Secured indicator	−46.26 (66.61)	−18.31 (46.28)
Covariates	No	Yes
Industry FE	Yes	Yes
Year FE	No	No
Observations	3,202	3,010
Adjusted R ²	0.06	0.12
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Table A.V: Marblegate effect in the full-period sample. This table re-estimates the baseline difference-in-differences specification using the full sample period, including observations after 2016. The dependent variable is the loan premium in basis points. M_t is an indicator equal to one for loans originated in 2015 or 2016, and $B_{i,t}$ is an indicator equal to one if the issuer's bond holdings scaled by book assets exceed the sample median of 0.26. The coefficient of interest is the interaction $M_t \times B_{i,t}$. Sector fixed effects are based on 2-digit SIC sectors, and standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>	
	Loan premium	
	(1)	(2)
Marblegate	1.46 (19.35)	1.46 (19.35)
Bond intensive	42.07*** (10.48)	42.07*** (10.48)
Marblegate*Bond intensive	-64.31*** (16.09)	-64.31*** (16.09)
Covariates	No	Yes
Industry FE	Yes	Yes
Year FE	No	No
Observations	4,922	4,922
Adjusted R ²	0.06	0.06

Note: *p<0.1; **p<0.05; ***p<0.01

Table A.VI: Marblegate effect in the term-loan-only sample. This table re-estimates the baseline difference-in-differences specification in the term-loan-only subsample. The dependent variable is the loan premium in basis points. M_t is an indicator equal to one for loans originated in 2015 or 2016, and $B_{i,t}$ is an indicator equal to one if the issuer's bond holdings scaled by book assets exceed the relevant sample median. The coefficient of interest is the interaction $M_t \times B_{i,t}$. Sector fixed effects are based on 2-digit SIC sectors, and standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>	
	Loan premium	
	(1)	(2)
Marblegate	36.67 (25.16)	44.21** (21.65)
Bond intensive	85.45** (36.56)	126.69*** (33.99)
Marblegate*Bond intensive	−69.48* (39.39)	−65.03** (32.40)
Covariates	No	Yes
Industry FE	Yes	Yes
Year FE	No	No
Observations	1,126	1,052
Adjusted R ²	0.08	0.10
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Table A.VII: Oil crisis and the Marblegate effect. This table examines whether the estimated Marblegate effect is confounded by the 2014–2016 oil crisis. The dependent variable is the loan premium in basis points, and the sample is restricted to loans originated through 2016. In column (1), I augment the baseline specification with an interaction between the Marblegate indicator and an oil-exposure indicator. In column (2), I re-estimate the baseline specification after excluding oil-exposed firms. $B_{i,t}$ is an indicator equal to one if the issuer’s bond holdings scaled by book asset exceed the sample median. Sector fixed effects are based on 2-digit SIC sectors, and standard errors are clustered at the issuer-firm and year levels.

	<i>Dependent variable:</i>	
	Loan premium	
	Include oil firms	Exclude oil firms
	(1)	(2)
Marblegate	39.46*** (14.99)	31.25** (15.68)
Bond intensive	186.79*** (26.85)	176.76*** (25.59)
Marblegate*Bond intensive	−70.42*** (20.83)	−50.65** (21.73)
Marblegate*Oil Firm	−363.15** (153.95)	
Industry FE	Yes	Yes
Year FE	No	No
Firm Control	Yes	Yes
Observations	3,010	2,803
Adjusted R ²	0.16	0.12

Note: *p<0.1; **p<0.05; ***p<0.01

Table A.VIII: CDS spread, loan premium, and subsequent rating-score change. This table regresses the change in the issuer's numeric credit rating score within five years of loan origination on the loan premium and, in column (2), the firm's 5-year CDS spread. The CDS spread is measured in percentage points. All specifications include the issuer's initial rating, industry fixed effects, and year fixed effects. The sample is restricted to observations for which CDS data are available.

	<i>Dependent variable:</i>	
	Rating change (score) within 5 years	
	(1)	(2)
Loan premium	−0.069*** (0.021)	−0.068*** (0.021)
Initial rating	−0.129*** (0.025)	−0.133*** (0.026)
CDS spread (5Y)		0.005 (0.005)
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	571	571
Adjusted R ²	0.227	0.227
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01		